GMODet: A Real-Time Detector for Ground-Moving Objects in Optical Remote Sensing Images With Regional Awareness and Semantic–Spatial Progressive Interaction

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***Abstract*— Military conflicts have a significant impact on national security and the ecological environment. Effective detection methods for ground-moving objects in complex scenarios can be further utilized to assess damage and provide recommendations for security and environmental restoration. Current ground-moving object detection in optical remote sensing imagery struggles with balancing detection accuracy and real-time performance, hindering timely threat assessment. To address this, the study proposes ground-moving object detector (GMODet), a real-time detection method incorporating region awareness and semantic–spatial interaction to enhance the detection of partially occluded and fine-grained objects in complex environments. The framework includes three modules: the region awareness module (RAW), cross-scale context-aware feature aggregator (CCFA), and semantic–spatial progressive interaction module (SPIM), focusing on extracting discriminative features for contextual, multiscale, and semantic–spatial information. A new dataset, ground-based moving object dataset (GMOD), is constructed with four object types and high scene complexity, alongside experiments on the publicly available military vehicle remote sensing dataset (MVRSD). GMODet achieves the state-of-the-art performance, with mAP_{50}, mAP_{75}, and mAP scores of 65.5%, 48.5%, and 42.3% on the GMOD, outperforming the second-best results by 1.9%, 5.1%, and 1.5%, respectively. On the MVRSD, it achieves mAP_{50}, mAP_{75}, and mAP scores of 88.2%, 75.2%, and 61.7%, respectively. Notably, with an inference time of just 25 s on large-scale images (9152 × 9152 pixels), GMODet showcases outstanding accuracy, speed, robustness, and generalization in ground-moving object detection.**

***Index Terms*— Ground-moving objects, optical remote sensing images, real time, regional awareness, semantic–spatial progressive interaction.**

I. Introduction

BEFORE and after the occurrence of military conflicts, ground-moving objects can pose potential threats to national security and the ecological environment. Remote sensing detection methods enable the rapid and intelligent identification of ground-moving objects, which is of critical importance for real-time dynamic object monitoring tasks. The acquired information, such as object categories and behavioral states, can be directly applied to decision-making and analysis in various fields, including disaster detection, emergency rescue, traffic management, national security, and environmental protection [1], [2], [3], [4]. By utilizing pre-conflict publicly available remote sensing data in conjunction with object detection technology for assessment, potential threats to national security and the ecological environment can be mitigated. Alternatively, postconflict remote sensing data combined with object detection technology can be used to estimate the extent of threats and environmental damage, providing decision-makers with recommendations for national security and environmental reconstruction strategies.

Remote sensing object detection plays a crucial role in ensuring national defense and ecological security. Inadequate detection can result in missed ground-moving objects, leading to security threats and ecological damage [5], [6]. Real-time monitoring of these objects in conflict zones is essential for effective response strategies and for safeguarding national defense and the environment. With advancements in Earth observation technology, remote sensing now allows for more detailed detection, even in complex scenarios. However, challenges, such as occlusions from trees, smoke, and lighting conditions, can hinder the detection of small, weak, and distant objects [7], [8], [9]. Overcoming these challenges is vital for rapid threat assessment and preventing further harm to national security and the environment.

Recent advancements in remote sensing object detection have been driven by the integration of artificial intelligence (AI) technologies. Two-stage methods, such as faster R-convolutional neural network (CNN) [10], improve accuracy by generating candidate regions and refining classifications, while single-stage networks, such as YOLO (e.g., YOLOv5, YOLOv7, YOLOv8) and their optimized versions, offer faster detection, making them ideal for real-time applications [11], [12], [13]. Transformer-based models [14], such as Swin Transformer [15], [16] and DEtection Transformer (DETR) [17], inparticular the DETR model, use a multihead

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self-attention mechanism, eliminating the need for region proposal generation and nonmaximum suppression (NMS) [18], simplifying postprocessing. Enhanced versions, such as efficient-DETR [19] and RT-DETR [20], optimize computational efficiency and feature extraction. Moreover, AGMF-Net [21], GLSANet [22], and TMAFNet [23] address challenges in detecting small objects and handling complex backgrounds in remote sensing images. These models employ multiscale fusion, global context understanding, and background noise reduction. However, balancing computational efficiency with high performance across diverse object scales remains an ongoing challenge.

Although current methods show promising results in detecting moving objects in remote sensing imagery, they still encounter significant challenges when applied to fine-grained ground-moving object detection in complex scenarios [24], [25]. This is especially critical in military contexts, where detecting objects against a cluttered background is crucial for national security and environmental threat assessment. Numerous studies have tackled the challenge of background noise in remote sensing images by developing more advanced network architectures. These include the top–down aggregation network, which enhances shallow feature maps by integrating deeper semantic information to detect armored objects [26], networks that combine style transfer with domain adversarial learning for few-shot vehicle detection in military settings [27], and a transformer-based module integrated into a single-stage detection model, which effectively captures global contextual information, thereby enhancing performance in military vehicle (MV) detection [28]. While these approaches have led to improvements in detection accuracy, their large parameter sizes and high computational demands limit their practicality in resource-constrained environments, such as unmanned aerial vehicles (UAVs) or satellites. Additionally, the majority of existing datasets are based on simulated scenarios, often lacking real-world data. This discrepancy means that they do not adequately capture the complexities of real-world conditions, such as smoke, occlusion, and dynamic changes in the environment. Another pressing challenge lies in the small interclass differences and large intraclass variations that complicate ground-moving object detection. For example, occlusions and obstructions can cause objects from different categories to appear similar, making it harder for models to distinguish them. These challenges not only undermine the accuracy of ground-moving object detection but also pose direct threats to national security and the ecological environment, potentially causing prolonged and unnecessary harm. Addressing these challenges is therefore of critical importance for advancing remote sensing applications in the detection of ground-moving objects.

To address the aforementioned challenges, this study proposes a real-time detection method for ground-moving objects in optical remote sensing images, supported by deep learning techniques that incorporate region awareness and semantic–spatial progressive interaction. This method, termed ground-moving object detector (GMODet), aims to strike a balance between detection accuracy and real-time performance, thereby contributing to the maintenance of national defense security and the stability of the ecological environment. The foundational framework of GMODet relies on DETR, which minimizes the time consumed in postprocessing operations and achieves a fully end-to-end detection system. To enhance the representation capability of object features in complex scenes of high-resolution optical remote sensing images, we propose a region awareness module (RAW). This module is designed to bolster the representation of global information from distant ground-moving objects and local information from adjacent pixels, as well as to implement diversified high-level semantic feature awareness strategies, thereby improving the rapid localization of potential regions associated with the same category of objects. Furthermore, the cross-scale context-aware feature aggregator (CCFA) is devised to aggregate valuable information contained in adjacent feature maps across different scales, guiding the network's focus on multiscale features. Finally, the semantic–spatial progressive interaction module (SPIM) establishes a bridge between semantic and spatial information, enhancing guided learning for different categories of objects through the integration of coarse semantic information and shallow spatial features. This is particularly crucial for distinguishing between similar objects and backgrounds, as well as for fine-grained identification of ground-moving objects.

1) This study innovatively proposes a real-time detection framework for ground-moving objects in real-world scenarios. The framework consists of three main modules: RAW, CCFA, and SPIM. The interaction among these modules addresses the challenge of balancing detection accuracy and real-time performance for fine-grained ground-moving object detection in optical high-resolution remote sensing imagery under complex environmental conditions.
2) A lightweight context-aware weight aggregator (CWA) structure has been designed for feature fusion in the proposed real-time object detection network, enhancing the aggregation of valuable information.
3) A dataset for fine-grained detection of aerial ground-moving objects in realistic complex scenarios has been established. This dataset includes various scenes, such as plains, rural areas, and urban environments, featuring large capture angles and containing numerous extremely distant, small, and partially occluded objects. The proposed method has been validated on this dataset and an open-source satellite ground vehicle object dataset, demonstrating its superiority.

II. Related Work

Traditional moving object detection primarily focuses on utilizing digital image processing techniques to manually construct shallow features related to texture and geometry for the recognition of specific types of objects [29], [30]. These models have gradually been supplanted by machine learning methods due to the cumbersome feature construction processes, low detection rates, and time-consuming nature of traditional approaches. Conventional machine learning techniques, such as support vector machines (SVMs), decision

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trees (DTs), and logistic regression (LR), can be employed for moving object detection in more complex environments [31]. However, their limited feature construction and insufficient learning capabilities result in subpar performance in moving object detection when faced with complex scene transfer. With the continuous leap in deep learning technologies, a variety of moving object detection methods utilizing high-resolution optical remote sensing images, synthetic aperture radar (SAR) images, and infrared images have been proposed, demonstrating widespread applicability. Optical remote sensing images are frequently used for fine-grained detection of moving objects due to their diverse acquisition methods and the richness of object texture and shape information.

A. Object Detection in Remote Sensing Images

Remote sensing imagery often presents complex scene distributions and uneven object scales. Compounding this challenge, objects within the same category can exhibit significant differences in feature representation, whereas objects from different categories may display high degrees of similarity in their feature expressions [32]. These characteristics pose considerable challenges for applying deep learning-based object detection methods to remote sensing images, making it difficult to achieve superior detection performance directly. To address these issues, researchers have focused on two main aspects: diversifying training strategies and enhancing the ability to handle complex scenes. Through in-depth exploration of these areas, significant progress has been made, overcoming many of the difficulties and achieving notable research outcomes.

1) Diversified Training Strategies: Early deep learning-based remote sensing object detection technologies primarily focused on approaches, such as faster region-based CNN (RCNN), which involves generating proposal regions before performing localization and classification, and the YOLO series, which combines localization and classification processes concurrently. Subsequently, the advent of anchor-free object detection methods, exemplified by CenterNet [33] and FCOS [34], addressed the problem of excessive computational complexity due to numerous preset anchors, resulting in a significant reduction in computation time. Additionally, due to the scarcity of samples in specific remote sensing domains, research into few-shot object detection [35], [36] and weakly supervised object detection [37], [38], based on meta-learning approaches, has been introduced. Similarly, despite the abundance of remote sensing imagery, the associated labeling costs have increased. To mitigate this, a remote sensing training strategy based on "point-faux box-refined box" has been proposed [39]. This strategy significantly reduces manual annotation costs through point-based labeling and demonstrates excellent object detection performance in complex scenarios. Moreover, while pretraining models using datasets, such as ImageNet [40], is common in general fields to accelerate the training process, analogous efforts in remote sensing have emerged. These approaches leverage general knowledge to enhance the pretraining process of remote sensing images, thus ensuring model performance and facilitating real-time online training for new categories [41]. On the other hand, the accumulation of multimodal remote sensing data has made the training of foundational remote sensing models feasible. Models, such as RingMo [42] and Skysense [43], provide robust transfer learning capabilities for object detection across various scenarios.

2) Complex Scene Handling Capabilities: Currently, to address the deficiencies in remote sensing image object detection caused by significant variations in object scales [44], [45], several strategies have been proposed, including the integration of multimodal data for information complementarity [46], scale enhancement strategies [47], [48], [49], [50], and optimization-based attention mechanisms [51], [52], [53]. These approaches have achieved high detection performance. For complex and densely packed object detection problems, several advanced methods designed for detecting objects in arbitrary orientations have been introduced [54]. These methods ensure comprehensive coverage and bounding of objects, thereby reducing both missed detections and false detections. In large-scale remote sensing imagery for ship detection, an improved YOLO-based rotation detector, which eliminates the need for angle regression, has demonstrated excellent detection performance and significantly improved speed compared to the state-of-the-art algorithms [55]. Zhan et al. [56], addressing challenges, such as anchor-point dependence, low object orientation accuracy, and high computational complexity for arbitrary orientation detection, proposed a method that leverages inherent image features to compute object orientation and predict size, thereby directly generating accurate directed bounding boxes. This method, combined with a multitask loss function, has achieved fine-grained ship detection tasks in real marine complex environments. The small object problem remains a prominent issue in remote sensing object detection due to weak feature representation and susceptibility to occlusion. Effective strategies to mitigate false positives (FPs) and achieve superior small object detection performance include learning-based image super-resolution models [57], [58], class imbalance loss function-assisted convergence models [59], and CNN + transformer-based global and local information optimization models [60]. Additionally, researchers have conducted extensive and intriguing studies on spatial distribution differences of objects [61], [62], model convergence issues [63], [64], and complex objects (occluded and composite objects) [65], [66].

B. Military Object Detection

The detection of military objects is a key factor in assessing potential threats to national defense security and the ecological environment. It constitutes an important area of research in complex ground object detection. Therefore, it is necessary to provide a comprehensive overview of the current state of research in this field. They are susceptible to environmental interference, frequently obscured by factors, such as fog, smoke, and foliage. Consequently, military objects in remote sensing imagery exhibit complex and weakly represented features. Therefore, detecting military objects based on remote sensing imagery presents a highly challenging task.

1) Computation Efficiency and Model Optimization: In practical applications of military target detection, there

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are stringent requirements for computation efficiency and model optimization, driven by the need for efficient models in resource-constrained environments and the necessity to enhance detection accuracy and robustness. Regarding computational efficiency, researchers have reduced computational complexity through lightweight network designs. For instance, replacing the backbone network of YOLOv5 with MobileNetV3 significantly reduced model parameters while improving deployment performance [67], [68]. Additionally, optimizing multiscale feature extraction layers and refining loss functions have proven critical for enhancing detection performance [69]. In scenarios with limited data, challenges, such as data scarcity and class imbalance, have been addressed using techniques, such as spectral modeling algorithms and independent component analysis on hyperspectral imagery, which improve the generalization capability of models under data-deficient conditions [70], [71], [72], [73]. Beyond that, federated meta-learning and adversarial domain adaptation techniques have been adopted to enhance the performance of models trained on small datasets [74], [75]. Frameworks that utilize advanced large-scale language models for few-shot military object detection, in combination with diverse feature encoding methods, have demonstrated significant improvements in detection capabilities under low-data conditions [76], [77], [78].

2) Feature Extraction and Fusion Capabilities: Feature extraction and fusion capabilities are core challenges in military target detection, particularly in complex environments that demand more precise feature representation and robust performance. Researchers have proposed various strategies to enhance feature extraction capabilities. For instance, combining the EfficientDet model with generative adversarial networks (GANs) optimizes feature representations through adversarial learning, enabling the model to capture more discriminative features [79]. Similarly, coupling visual attention mechanisms with Gabor region proposal subnetworks has significantly improved the expression of fine-grained features [80]. In remote sensing imagery, using EdgeBoxes technology alongside superresolution methods to refine and filter object bounding boxes has enhanced the precision of feature extraction [81]. Moreover, to address the challenges of oblique perspectives in complex environments, researchers have developed models combining the local feature extraction capabilities of CNNs with the global feature extraction strengths of transformers, achieving improved detection accuracy [82]. Regarding camouflage and noise processing, researchers have further optimized feature extraction and fusion strategies. For instance, the YOLOv4-Faster network, utilizing a multiscale enhancement strategy, effectively addresses complex noise environments [83].

Simultaneously, the development of camouflage object detection models has improved detection accuracy in challenging environments [84]. Additionally, deformable DETR models incorporating reverse feature feedforward networks and multistage query feature fusion modules have significantly enhanced the detection of camouflaged objects [85]. These advancements collectively lay a solid foundation for applying military object detection in complex environments.

In conclusion, although remote sensing image object detection technology has played a significant role in various applied research areas, it still faces considerable challenges in detecting different fine-grained ground-moving objects. For instance, the difficulty in detecting partially hidden or occluded objects and the lack of real-time performance of object detection remain major limiting factors.

III. Methodology

A. Overall Network Framework

We propose a real-time ground-moving object detection framework, GMODet, as illustrated in Fig. 1, which adopts the RT-DETR architecture for end-to-end real-time processing. The accurate object detection results can inform assessments of potential threats to national defense and ecological environments. The environment faced by ground-moving objects is complex, and it is necessary to design relevant feature enhancement modules to alleviate the defect of insufficient detection capability caused by weak feature expression of ground-moving objects in remote sensing images. Innovatively, we have designed three feature enhancement blocks—RAM, CCFA, and SPIM—during the encoder phase to improve the network's discriminative power for ground-moving objects.

Feature extraction is a critical component of ground-moving object detection. The original image $\boldsymbol{I} \in \mathbb{R}^{H_I \times W_I \times 3}$ (where H_I and W_I denote the height and width of the original image) undergoes preprocessing steps, such as image flipping, scaling, and illumination adjustment before being fed into the backbone network for encoding. In this study, ResNet50 is selected as the backbone network. Consequently, the multiscale features ***S1***–***S5*** output dimensions are represented as $\boldsymbol{S_i} \in \mathbb{R}^{H_I/s_i \times W_I/s_i \times C_i}$, where $i \in [1], [5]$; s_i denotes the downsampling factor, which is an integer multiple of 2 with a maximum value of 32; and C_i represents the number of channels in the multiscale features, with $C \in \{128, 256, 512, 1024, 2048\}$.

After feature extraction through the backbone, the representation lacks discriminative features for the objects. The RAM is designed to integrate global and local contextual information from surrounding pixels, as well as to provide diverse scale feature representations. Notably, the number of feature channels in ***S1***–***S5*** is uniformly set to 256 before entering the Encoder. This configuration aims to reduce the computational parameter count while maintaining multiscale features. The input to RAM is denoted as $\boldsymbol{S5} \in \mathbb{R}^{H_I/32 \times W_I/32 \times 256}$. Within the internal structure of RAM, two branches—the global region awareness unit (GRAU) and the local region awareness unit (LRAU)—are employed to enhance the global and local contextual information for different objects. Meanwhile, the diverse scale awareness unit (DSAU) utilizes the high-level semantic information from ***S5*** and combines three distinct convolutional blocks to enhance the diversity of spatial multiscale feature representations at a higher level. Ultimately, a CWA fusion block integrates global information, local information, and diverse spatial scale features, thereby forming an enhanced region awareness capability.

RAM and CCFA tend to enhance high-level encoded features ***S5*** and their adjacent scales but lack focus on

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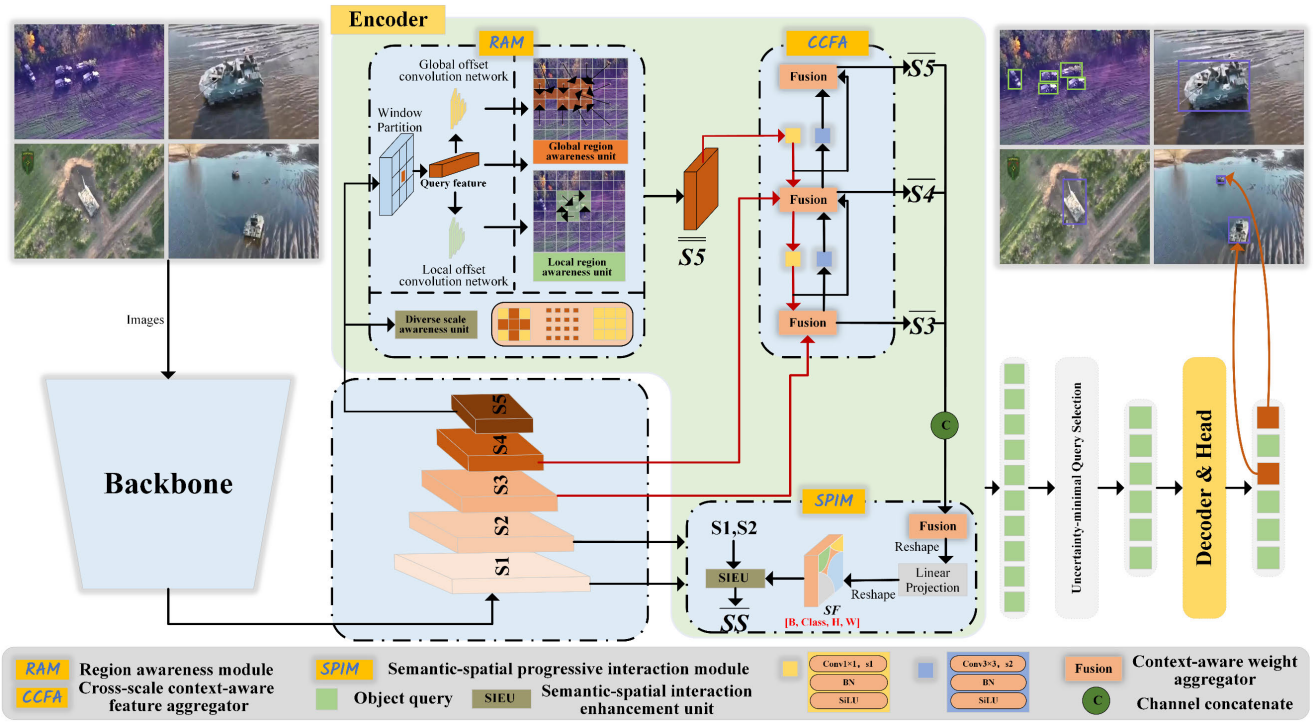


Fig. 1. Overall framework of GMODet. It consists of three main structural components: backbone, encoder, and decoder. The multiscale features from the backbone are input into the encoder. Through the intermediate feature enhancement provided by the region awareness module (RAM), CCFA, and SPIM, more discriminative features are generated. These enhanced discriminative features are then selected through the uncertainty-minimal query selection mechanism, which chooses a fixed number of encoder features as the initial object queries for the decoder. Finally, these object queries are fed into the decoder, which incorporates auxiliary prediction heads for iterative optimization, resulting in accurate predictions of object categories and bounding boxes.

discriminative features among different classes. Therefore, we input the multiscale aggregated features $\overline{\boldsymbol{S3}}$, $\overline{\boldsymbol{S4}}$, and $\overline{\boldsymbol{S5}}$ from CCFA into the designed SPIM, enabling the extraction of coarse semantic features $\boldsymbol{SF} \in \mathbb{R}^{H_I/4\times W_I/4\times \text{Class}}$ and combining them with shallow scale features from the backbone to progressively refine the spatial–semantic feature representation. The obtained spatial–semantic features $\overline{\boldsymbol{SS}} \in \mathbb{R}^{H_I/4\times W_I/4\times 256}$ are integrated with $\overline{\boldsymbol{S3}}$, $\overline{\boldsymbol{S4}}$, and $\overline{\boldsymbol{S5}}$ to serve as discriminative inputs for the uncertainty-minimal query selection structure [20], which selects a fixed number of encoded features as the initial object queries for the decoder. These object queries are then fed into a decoder equipped with auxiliary prediction heads for iterative optimization, resulting in accurate classifications of object types and bounding box predictions.

B. Region Awareness Module

In real-world scenarios, the distribution patterns of fine-grained ground-moving objects in remote sensing imagery are complex and diverse and are not fixed. Conventional network architectures typically do not focus on the differences in the distribution areas of various objects and lack discriminative features for fine-grained object detection. In other words, distant pixels and surrounding neighboring pixels collaboratively enhance the feature representation of ground-moving objects, which can improve the network structure's ability to learn contextual information, thereby assisting in object recognition. We need to establish a relationship between distant and neighboring pixel information to form both global and local discriminative features. Moreover, the scale effect of objects within the same category is particularly pronounced in remote sensing imagery. Consequently, designing a strategy that perceives diverse object scales within the same feature map can strengthen the recognition of objects belonging to the same category.

As shown in Fig. 2, the three feature enhancement units in our proposed RAM work synergistically to enhance performance. The inputs to the GRAU and LRAU are $\boldsymbol{S5} \in \mathbb{R}^{H_I/32\times W_I/32\times 256}$. Through a window partitioning operation, the feature map $\boldsymbol{S5}$ is divided into $(H_f/W_s) \times (W_f/W_s)$ windows, each of size $W_s \times W_s$. Here, H_f and W_f denote the dimensions of the feature map $\boldsymbol{S5}$, with the conditions $W_s < H_f$ and $W_s < W_f$. The process of window partitioning is illustrated in (1). Subsequently, self-attention needs to be executed within the range of each window to enhance the focus on object features, which also helps balance computational costs

$$f^{(1)}, f^{(2)}, \ldots, f^{(w)} = \text{WindowPartition}(\boldsymbol{S5}) \tag{1}$$

where $f^{(w)}$ representing the feature map of the wth window after segmentation.

1) GRAU: Specifically, for GRAU, it is essential to set a fixed reference point $f_g^r \in \mathbb{R}^{H_r\times W_r}$ within each window. The initialization value f_g^r is defined as a series of 2-D uniformly distributed grids $\{(0,0),(0,S),(0,2S),\ldots,(H_r,W_r)\}$, with all values normalized to the range of $[-1, 1]$. Here, $H_r = \lfloor (H_f-1)/S \rfloor \times S, W_r = \lfloor (W_f-1)/S \rfloor \times S$, S represents the downsampling factor. In this study, S is set to 2, meaning

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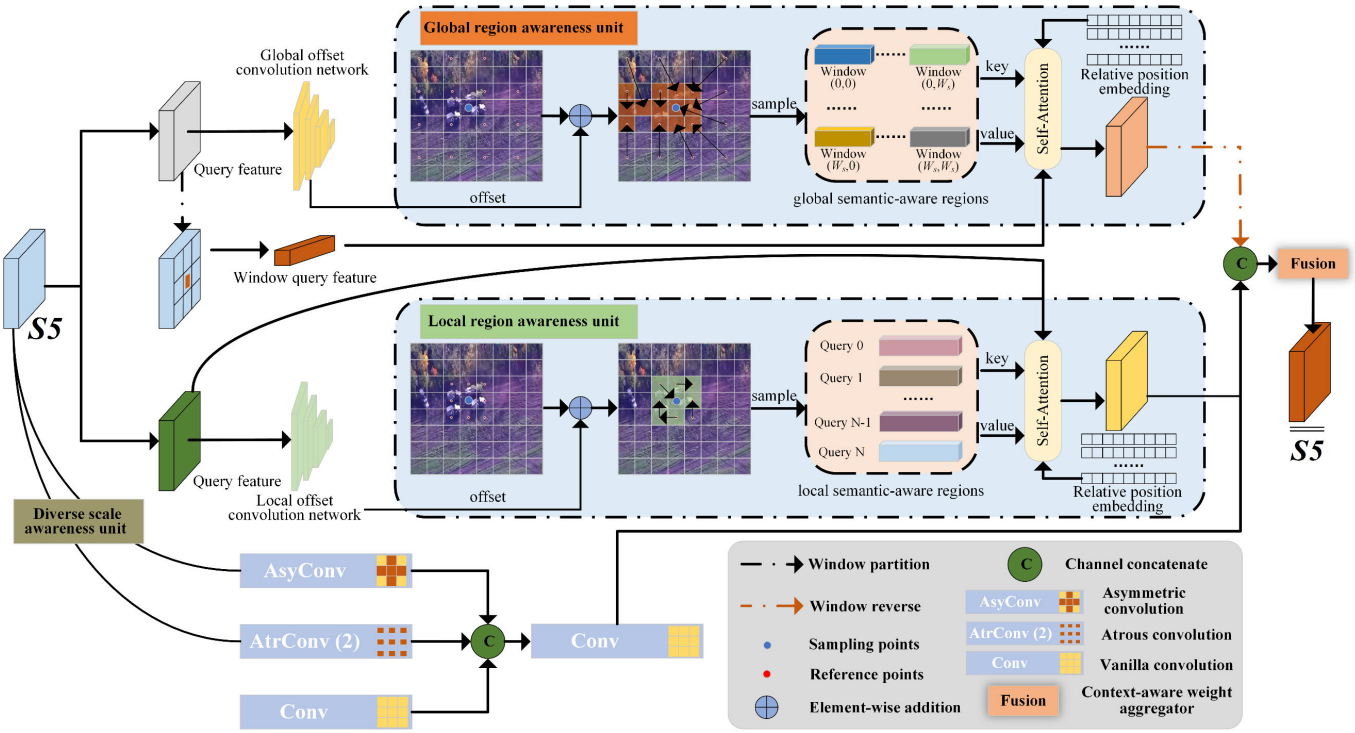


Fig. 2. Proposed region awareness module (RAM). The module consists of GRAU, LRAU, and DSAU.

that a pixel is sampled every two steps through a row-column cross-fixed sampling method. Our objective is to generate a series of corrected sampling points by adding the offsets $\Delta f_g^{(w)}$ and f_g^r produced by the global offset convolution network. It is noteworthy that the query features of self-attention within the same window share the offsets extracted by the offset network. The offset $\Delta f_g^{(w)}$ for each window is determined as described in (2). Subsequently, we utilize irregular bilinear interpolation $\psi(\cdot)$ to magnify these sampled point features, thereby generating global semantic-aware regions $\overline{f}_g^{(w)}$, as represented by (3)

$$q^{(w)} = W_q f^{(w)}, \quad \Delta f_g^{(w)} = \varepsilon_g\left(q^{(w)}\right) \tag{2}$$

$$\overline{f}_g^{(w)} = \psi\left(f, f_g^r + \Delta f_g^{(w)}\right) \tag{3}$$

where f denotes the feature map ***S5***, and its dimensions are $H_f \times W_f$. W_q represents the projection weight matrix of the query feature $q^{(w)}$ under the window feature $f^{(w)}$. $\varepsilon_g(\cdot)$ denotes the global offset convolution network.

The generated feature $\overline{f}_g^{(w)}$ is projected as key and value features, which are then fed into the self-attention mechanism module. In this study, the self-attention module employs relative position embeddings to maintain the spatial positional relationships of image features. For the input features $f \in \mathbb{R}^{H_f \times W_f}$, a relative position table $T \in \mathbb{R}^{(2H_f-1)\times(2W_f-1)}$ is constructed, which is learnable. In this study, $f \in \mathbb{R}^{H_f \times W_f}$ is ***S5***. The terms $2H_f - 1$ and $2W_f - 1$ represent the range of relative displacements in the vertical and horizontal directions, respectively. The relative displacement $\overline{R}$ at each sampling point is obtained by accumulating the fixed relative displacements R from each reference point and the adaptive displacements Δf generated by the offsets. Subsequently, the relative position embedding $P(f, \overline{f})$ can be sampled from T using $\overline{R}$. The specific computation process is detailed as follows:

$$\overline{R} = R + \Delta f \tag{4}$$

$$P(f, \overline{f}) = \psi(T, \overline{R}) \tag{5}$$

where P denotes the relative position encoding, while $\psi(\cdot)$ represents the bilinear interpolation operation.

Finally, the enhanced window features, which have undergone self-attention augmentation, are aggregated and projected back to the original feature size through the windows reverse operation, resulting in the enhanced global semantic information z_g. This process can be described as follows:

$$k^{(w)} = W_k \overline{f}_g^{(w)}, \quad v^{(w)} = W_v \overline{f}_g^{(w)} \tag{6}$$

$$z^{(w)} = \delta\left(\frac{q^{(w)}\left(k^{(w)}\right)^{\mathrm{T}}}{\sqrt{d}} + P\left(f, \overline{f}_g\right)\right) v^{(w)} \tag{7}$$

$$z_g = \mathrm{WindowReverse}\left(z^{(1)}, z^{(2)}, \ldots, z^{(w)}\right) \tag{8}$$

where W_k and W_v are the weights that are continually learned during the training process. $k^{(w)}$ and $v^{(w)}$ represent the key/value sets of the wth window, with d denoting the feature dimension. δ refers to the softmax function.

2) LRAU: As illustrated in Fig. 2, the structure of LRAU is similar to that of GRAU. Unlike GRAU, LRAU does not share the same offsets within the window; instead, each query feature is assigned a unique offset Δf_l. Furthermore, the reference point settings in LRAU take into account the similarity characteristics of neighboring pixels. Consequently, we initialize the reference point f_l^r with its eight neighboring pixels along with its own pixel point, represented as $\{(0, -1), (-1, -1), (-1, 0), (-1, 1), (0, 1), (1, 1), (1, 0), (1, -1),$

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(0, 0)}. The computational process of LRAU is detailed as follows:

$$q = W_q f, \quad \Delta f_l = \varepsilon_l(q) \tag{9}$$

$$\overline{f}_l = \psi\left(f, f_l^r + \Delta f_l\right) \tag{10}$$

$$k = W_k \overline{f}_l, \quad v = W_v \overline{f}_l \tag{11}$$

$$z_l = \delta\left(\frac{qk^{\mathrm{T}}}{\sqrt{d}} + P\left(f, \overline{f}_l\right)\right) v \tag{12}$$

where the local offset convolution network is represented by $\varepsilon_l(\cdot)$. The parameters W_q, W_k, and W_v are projection weights that are continuously learned throughout the training process.

Additionally, it is necessary to elucidate the structural composition of the global and local offset convolution networks. The global offset convolution network is comprised of the following sequence: "Conv(3, 1, 0)-Conv(3, 1, 1)-AvgPool(2, 2, 0)-Conv(3, 2, 1)-Conv(1, 1, 1)," with each convolutional layer followed by a ReLU activation and batch normalization. Specifically, the first four convolutional layers and the pooling layer are utilized for feature downsampling, while the final convolutional layer is employed to project the features into the number of reference points. In contrast, the local offset convolution network consists of two convolutional layers: "Conv(3, 1, 1)-Conv(1, 1, 0)," which are used for feature smoothing and projection into the number of reference points.

3) DSAU: The design of DSAU aims to enhance the perception of diverse features within scales, facilitating the recognition of objects of the same category. As illustrated in the lower left corner of Fig. 2, we employ convolutional layers with three distinct kernel sizes and shapes to perform multilevel feature extraction on the input feature ***S5***. Specifically, we utilize a combination of asymmetric convolution [86], atrous convolution [87], and vanilla convolution operations to capture more valuable scale-invariant features. Notably, the dilation rate for the atrous convolution is set to 2. Furthermore, the settings for the asymmetric convolution are consistent with those outlined in [88]. Consequently, the entire computational process can be represented as follows:

$$z_f^D = \mathcal{X}_{\mathrm{ori}}\left(\mathrm{Concat}\left(\mathcal{X}_{\mathrm{asy}}(f), \mathcal{X}_{\mathrm{atr}}^2(f), \mathcal{X}_{\mathrm{ori}}(f)\right)\right) \tag{13}$$

where f denotes the feature map ***S5***, $\mathcal{X}_{\mathrm{asy}}(\cdot)$ represents the asymmetric convolution, $\mathcal{X}_{\mathrm{atr}}^2(\cdot)$ denotes the atrous convolution with a dilation rate of 2, $\mathcal{X}_{\mathrm{ori}}(\cdot)$ signifies vanilla convolution, and Concat refers to channel dimension fusion.

Ultimately, the output features $\overline{\overline{\boldsymbol{S5}}} \in \mathbb{R}^{H_I/32\times W_I/32\times 256}$ of RAM consist of the fusion of GRAU, LRAU, and DSAU features, which are adjusted in weight by the CWA, and Aggregator denoted as the CWA. A detailed description of the CWA will be provided in Section III-C. The feature aggregation process described above can be represented as follows:

$$\overline{\overline{\boldsymbol{S5}}} = \mathrm{Aggregator}\left(\mathrm{Concat}\left(z_g, z_l, z_f^D\right)\right). \tag{14}$$

C. Cross-Scale Context-Aware Feature Aggregator

Fig. 1 illustrates the architecture of CCFA. The inputs to CCFA consist of three multiscale features: the processed output from RAM, denoted as $\overline{\overline{\boldsymbol{S5}}}$, and the features ***S4*** $\in \mathbb{R}^{H_I/16\times W_I/16\times 256}$ and ***S3*** $\in \mathbb{R}^{H_I/8\times W_I/8\times 256}$ obtained from the backbone network. For the same object, the shape and size discrepancies between adjacent feature maps can be significant; however, there still exists a close relationship between them. Thus, this structure is employed to aggregate the valuable information contained in the different scales of adjacent feature maps. The multiscale fusion path within this module incorporates three fusion blocks, which are the crucial components of CCFA. As depicted in Fig. 3, the fusion block first enhances features in the frequency domain and then adaptively reweights the input features to maximize the expressive capability of valuable information for each feature. The CWA fusion block is utilized multiple times within the network structure, but the frequency enhancement unit (FEU) is only incorporated in CCFA, with its presence removed in the other instances. The CWA can be mathematically represented by an equation. The calculations for frequency-domain enhancement are outlined as follows:

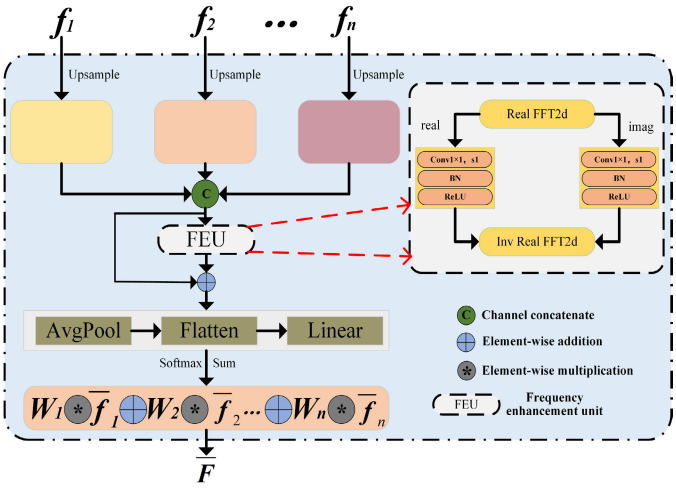


Fig. 3. CWA. Utilizing frequency-domain enhancement processing, this module adaptively reweights the input features to output aggregated features that contain valuable information.

$$\text{Real FFT2d: } \mathbb{R}^{H\times W\times C} \to \mathbb{C}^{H\times W/2\times C} \tag{15}$$

$$\text{ReLU} \circ \text{BN} \circ \text{Conv1} \times 1\text{: } \mathbb{R}^{H\times W/2\times 2C} \to \mathbb{R}^{H\times W/2\times 2C} \tag{16}$$

$$\text{RealToComplex: } \mathbb{R}^{H\times W/2\times 2C} \to \mathbb{C}^{H\times W/2\times C} \tag{17}$$

$$\text{Inverse Real FFT2d: } \mathbb{C}^{H\times W/2\times C} \to \mathbb{R}^{H\times W\times C}. \tag{18}$$

The computational process outlined in (15)–(18) involves transforming the input features into the frequency domain, where the real and imaginary parts of the frequency-domain features are separately enhanced through convolutional operations. Following this enhancement, the real and imaginary parts are fused together and then transformed back to the spatial domain. The features obtained from the inverse transformation are combined with the original input features using a residual block, and this combined feature set is fed into a multilayer perceptron to obtain learnable weights for each feature. The detailed calculation process is illustrated

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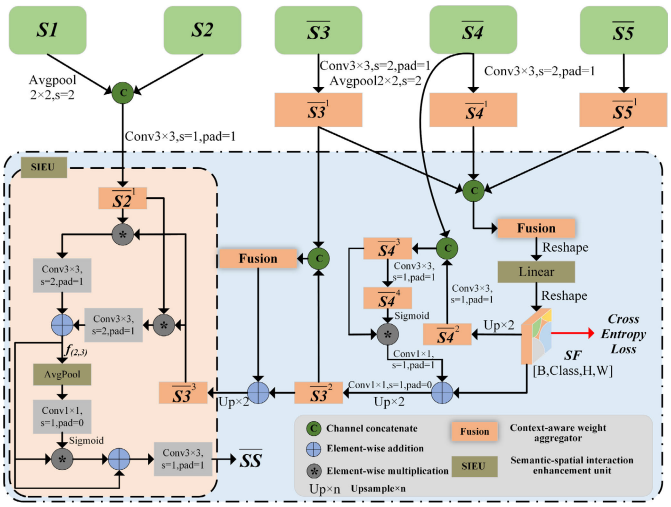


Fig. 4. Designed SPIM structure. This module takes multilevel, multiscale features as input and processes them through the SIEU, resulting in the semantic–spatial aggregated features $\overline{SS}$.

as follows:

$$\begin{aligned} \boldsymbol{W} &= \{\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_n\} \\ &= \text{softmax}(\boldsymbol{MLP}([\overline{f}_1, \overline{f}_2, \ldots, \overline{f}_n]) \end{aligned} \tag{19}$$

$$\overline{\boldsymbol{F}} = \sum_{i=1}^{n} \boldsymbol{w}_i \cdot \overline{f}_i \tag{20}$$

where $\overline{f}_i$ denotes the 1-D feature after average pooling (AvgPool) shown in Fig. 3, $i \in [1, n]$. ***MLP*** represents the operation of a multilayer perceptron, which, after applying the softmax function, yields the weight matrix $\boldsymbol{W}$. The symbol $\cdot$ represents matrix multiplication. Ultimately, the fused feature $\overline{\boldsymbol{F}}$ is obtained by summing the individual features along with their corresponding weights.

D. Semantic–Spatial Progressive Interaction Module

As shown in Fig. 4, we have designed a strategy called SPIM aimed at enhancing the differential feature recognition and identification accuracy of various object categories. Currently, most object detection algorithms tend to focus on the search and attention of semantic information, often neglecting the significance of shallow spatial information. The implicit interaction between semantic information and spatial features is more conducive to generating valuable fine-grained discriminative features. The inputs to SPIM consist of the multiscale high-level features $\overline{\boldsymbol{S3}}$, $\overline{\boldsymbol{S4}}$, and $\overline{\boldsymbol{S5}}$ enhanced by CCFA, along with the features ***S1*** and ***S2*** obtained from the backbone network.

The multiscale features $\overline{\boldsymbol{S3}}$, $\overline{\boldsymbol{S4}}$, and $\overline{\boldsymbol{S5}}$ are utilized to capture coarse semantic information. The detailed computational process is shown as follows:

$$\boldsymbol{SF} = \text{Linear}\left(\text{Aggregator}\left(\text{Concat}\left(\overline{\boldsymbol{S3}}^1, \overline{\boldsymbol{S4}}^1, \overline{\boldsymbol{S5}}^1\right)\right)\right) \tag{21}$$

$$\begin{aligned} \overline{\boldsymbol{S3}}^2 = \text{Conv1} \times 1\Big(\sigma\Big(\text{Conv3} \times 3\Big(\text{Concat}\Big(\overline{\boldsymbol{S4}}, \overline{\boldsymbol{S4}}^2\Big)\Big)\Big) \\ \cdot \text{Concat}\Big(\overline{\boldsymbol{S4}}, \overline{\boldsymbol{S4}}^2\Big)\Big) + \boldsymbol{SF}. \end{aligned} \tag{22}$$

The term Linear refers to a fully connected layer. The symbol $\cdot$ represents matrix multiplication, and σ refers to the sigmoid function. ***SF*** denotes the extracted coarse semantic information, which contains only class-related information and is used to guide the enhancement of fine-grained object recognition accuracy. The variable $\overline{\boldsymbol{S3}}^2$ represents an intermediate feature map that incorporates both semantic and spatial information.

More importantly, semantic–spatial interaction enhancement unit (SIEU) is a crucial component of the SPIM structure. Specifically, this study first performs dimension matching between ***S1*** and ***S2***, enabling concatenation along the channel dimension to capture more spatial information. Subsequently, the concatenated features undergo feature smoothing with a convolution kernel size of 3, resulting in feature map $\overline{\boldsymbol{S2}}^1$. This process can be represented as follows:

$$\overline{\boldsymbol{S2}}^1 = \text{Conv3} \times 3(\text{Concat}(\boldsymbol{S_1}, \boldsymbol{S_2})). \tag{23}$$

We designed an interactive connection strategy to ensure the effective interaction between spatial and semantic information, as well as the integration of valuable clues. The interaction process is formulated, as shown in (24). On this basis, a self-attention mechanism is employed to mine the most critical clues from the obtained interactive features $\boldsymbol{f}_{(\boldsymbol{2,3})}$, thereby enhancing the self-focus capability on the object features. This process is described in (25). The coupled feature map $\overline{\boldsymbol{SS}}$, which integrates semantic and spatial information, is obtained and input, together with multiscale features $\overline{\boldsymbol{S3}}$, $\overline{\boldsymbol{S4}}$, and $\overline{\boldsymbol{S5}}$, into the uncertainty-minimal query selection and decoder layers. Ultimately, the object information, including category and coordinate information, is returned as output

$$\boldsymbol{f}_{(\boldsymbol{2,3})} = \text{Conv3} \times 3\left(\overline{\boldsymbol{S2}}^1 \cdot \overline{\boldsymbol{S3}}^3\right) + \text{Conv3} \times 3\left(\overline{\boldsymbol{S3}}^3 \cdot \overline{\boldsymbol{S2}}^1\right) \tag{24}$$

$$\begin{aligned} \overline{\boldsymbol{SS}} = \text{Conv3} \times 3(\sigma(\text{Conv3} \times 3\left(\text{Avgpool}\left(\boldsymbol{f}_{(\boldsymbol{2,3})}\right)\right) \cdot \boldsymbol{f}_{(\boldsymbol{2,3})} \\ + \boldsymbol{f}_{(\boldsymbol{2,3})} \end{aligned} \tag{25}$$

where the term Avgpool refers to the global average pooling operation.

E. Hybrid Loss Constraints

In this study, a hybrid loss function was employed during the decoding stage, incorporating multiple loss constraints to ensure the stability and convergence of the training process. The specific loss function formulas are presented as follows:

$$\begin{aligned} \text{loss_vfl} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} \alpha \cdot p_{i,c}^{\gamma} \cdot (1 - y_{i,c}) + \text{IoU}(B_i, \hat{B}_i) \\ \cdot y_{i,c} \log(\sigma(z_{i,c})) \end{aligned} \tag{26}$$

$$\text{loss_bbox} = \frac{1}{N}\sum_{i=1}^{N} \|\hat{B}_i - B_i\|_1 \tag{27}$$

$$\text{loss_giou} = \frac{1}{N}\sum_{i=1}^{N}(1 - \text{GIoU}(B_i, \hat{B}_i) \tag{28}$$

$$\text{loss_ce} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c} \log(p_{i,c}) \tag{29}$$

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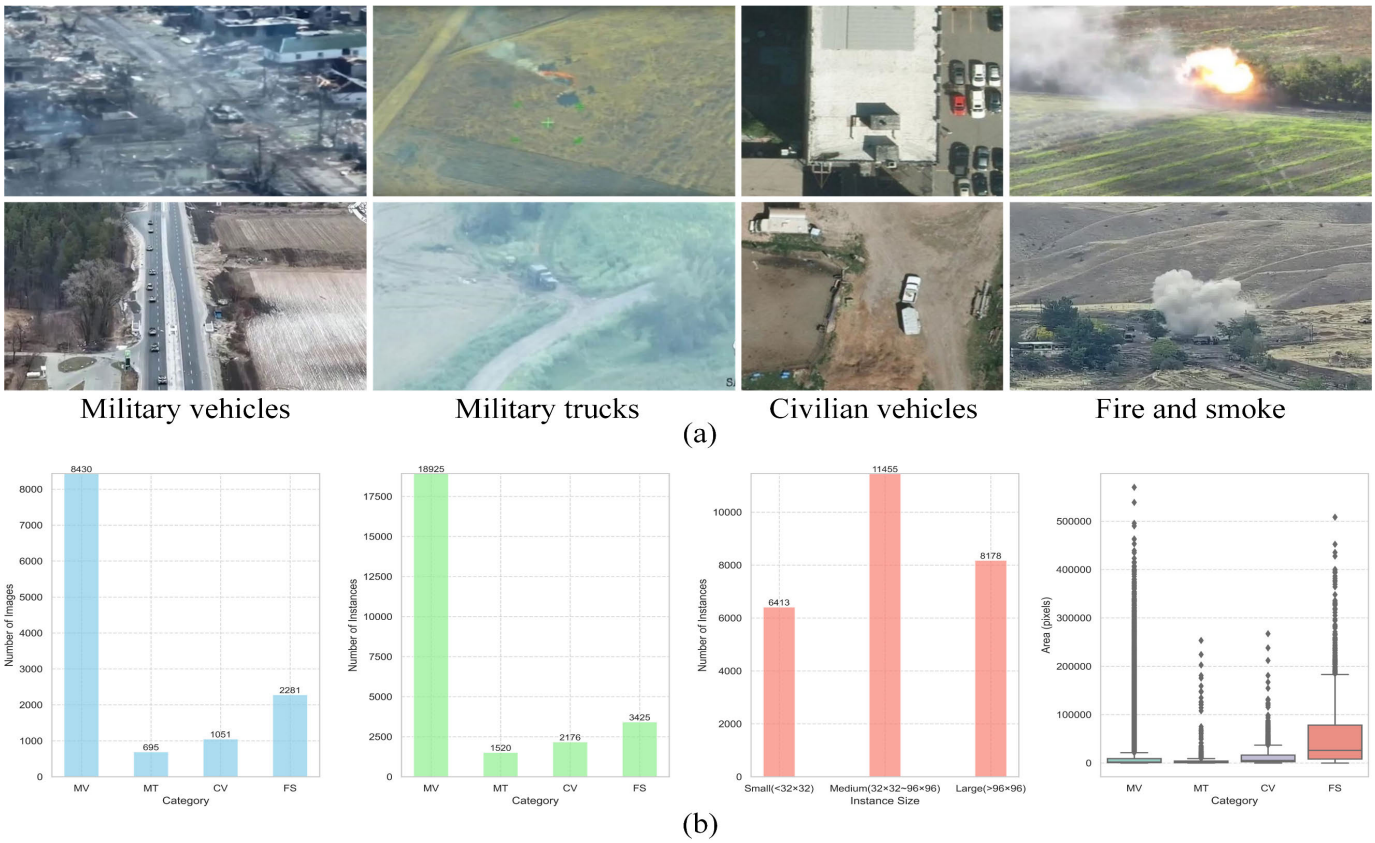


Fig. 5. Examples of various ground-moving objects in the GMOD and dataset statistics. (a) Dataset includes four categories: MVs, MTs, CVs, and FS. The images exhibit a complex and diverse backgrounds, making the object detection task challenging. (b) Detailed statistics of the dataset are presented, including the number of images for each object type, the number of instances for each object, statistics regarding instance sizes, and area size statistics for instances across various categories. It is observed that the dataset exhibits the characteristics of class imbalance.

$$\text{Hybrid_Loss} = \lambda_1 \cdot \text{loss_vfl} + \lambda_2 \cdot \text{loss_bbox} + \lambda_3 \cdot \text{loss_giou} + \lambda_4 \cdot \text{loss_ce} \tag{30}$$

where N represents the number of samples; α is a weighting factor; C denotes the number of categories; $y_{i,c}$ is the one-hot encoding of the object categories; σ represents the sigmoid function; γ is the focal parameter used to control the weighting of hard samples; $p_{i,c}$ indicates the model's predicted probability. Intersection over union (IoU) and generalized IoU (GIoU) refer to the IoU loss [89] and generalized IoU loss [90], respectively. B_i represents the coordinates of the predicted bounding box, while $\hat{B}_i$ represents the coordinates of the ground truth bounding box. $\lambda_1 - \lambda_4$ represent the weights of the respective loss functions, which are set to 1, 5, 2, and 1 in this study.

IV. Experiment and Analysis

A. Datasets and Evaluation Metrics

To evaluate the effectiveness and validity of the method proposed in this study, we conducted verification not only using the ground-based moving object dataset (GMOD), a dataset of UAV ground moving target detection in complex scenarios that we collected, but also tested its generalization capability on an open-source satellite imagery dataset. The relevant parameters for both datasets are shown in Table I.

TABLE I
Details of the Validation Data Used in the Experiments

Dataset	Numbers	Categories	Size	Resolution
GMOD	11,262	MVs, MTs, CVs, FS	~1,280×1,280 pixels	-
MVRSD	3,000	SMVs, LMVs, AFVs, MCVs, CVs	640×640 pixels	0.3 m

1) GMOD: Specifically, we collected images and video data containing ground-moving objects from drone monitoring conditions in real conflict zones extensively over recent years. After processing the relevant data, followed by manual annotation and expert verification, we obtained the final GMOD. Most images in the GMOD feature instances of moving ground objects captured by UAVs in different regions, annotated using horizontal bounding boxes. As shown in Fig. 5(a), the GMOD presents a complex background, with intricate scene distributions that encompass plain, rural, and urban environments. The displayed object instances illustrate the significant angles of inclination in UAV imagery, with numerous distant, very small, and occluded objects. Additionally, there is strong similarity between different categories of objects, while substantial variability exists within the same object category. The GMOD is classified into four categories: MVs, military trucks (MTs), civilian vehicles (CVs), and fire and

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smoke (FS). In this study, we accumulated annotations for 11 262 images, with the maximum image resolution reaching 1280 × 1280 pixels. Furthermore, as illustrated in Fig. 5(b), we annotated a total of 26 046 object instances. Quantitative statistical analysis indicates that the distribution of each category and the sizes of object instances are highly imbalanced, reflecting the challenges present in real-world scenarios and posing significant difficulties for object detection. In this study, we divided the GMOD into training, validation, and testing sets in a ratio of approximately 7:2:1.

2) MVRSD: This dataset is based on Google Earth data and comprises ground moving vehicle objects collected from over 40 typical scenarios across Asia, North America, and Europe. The geographical scenes primarily feature complex backgrounds, such as urban areas, plains, mountains, and deserts. There are five categories in total: small MVs (SMVs), large MVs (LMVs), armored fighting vehicles (AFVs), military construction vehicles (MCVs), and CVs. The dataset includes vehicle objects with varying degrees of camouflage and movement states, effectively reflecting the challenges in detecting ground-moving vehicles due to environmental camouflage, variations in shape and structure, and motion blur. A total of 3000 satellite images with a resolution of 0.3 m have been annotated, each sized 640 × 640 pixels. The total number of annotated object instances reaches 32 626. In addition, the dataset includes three images depicting densely distributed ground vehicle scenes on a large scale, which are used to validate the detection reliability under the conditions of substantial size. In this study, the distribution of the dataset follows that of the original paper's experimental design [91], with a training-to-testing ratio of 4:1, although 25% of the training set is allocated for validation.

3) Evaluation Metrics: To validate and assess the performance of the method proposed in this study, as well as various advanced approaches, we adopted standard evaluation metrics, including average precision (AP) ($AP_{50:95}$), AP at 50% IoU (AP_{50}), and AP at 75% IoU (AP_{75}), in accordance with the criteria set forth in [92]. Additionally, we utilized AP_{50} for the quantitative evaluation of detection performance across each object category. For the statistical analysis of model parameters and computational complexity, we employed overall parameters (Params) and Giga floating point operations (GFLOPs) as the statistical criteria. In terms of computational efficiency, frames per second (FPS) were also used to evaluate the performance of each method [93].

B. Experimental Details

1) Experimental Setup: All experiments in this study were conducted on an Ubuntu 20.04 system using a single NVIDIA RTX 3090 GPU with 24 GB of memory. The deep learning framework for our proposed method is based on PyTorch 2.0. During the training phase, we applied varying degrees of data augmentation to the dataset, including random brightness adjustments, random scaling, and horizontal flipping. For the two experimental datasets, we utilized multiscale images during the training phase. The original images were resized to a uniform dimension of 672 × 672 pixels, followed by multiscale resizing, resulting in four input scales of 480 × 480, 576 × 576, 672 × 672, and 768 × 768 pixels. During the inference phase, we used an input size of 672 × 672 pixels to balance inference speed and detection accuracy. In the training phase, we set the batch size to 10. The GMOD and military vehicle remote sensing dataset (MVRSD) were each trained for 200 epochs to ensure adequate convergence. The initial learning rate was set to 0.0001, and the AdamW optimizer was chosen with a weight decay of 0.0001.

2) Comparison Methods: To validate the superior performance of the proposed method, it is essential to compare it with the current state-of-the-art representative methods. We conducted comparisons from two perspectives: CNN-based models and transformer-based models. The CNN-based models examined in this study include WTHA-ViT [94], YOLOF [95], HANet [96], FSANet [97], TOOD [98], faster R-CNN [10], NWD-RKA [99], sparse R-CNN [100], YOLOV8-L [101], and PPYOLOE-L [102]. For transformer-based models, we included the most representative DETR [15] and its various variants, including conditional-DETR [103], DNTR [104], DAB-DETR [105], CO-DETR [106], DINO-Deformable-DETR (DINO-DETR) [107], Lite-DETR [108], and RT-DETR [20]. These networks have been widely applied and validated in object detection tasks in the fields of remote sensing and others, achieving outstanding results. To ensure fairness in the experiments, we maintained the original parameter settings of these network architectures.

C. Experiments on GMOD Task

The quantitative evaluation metrics in Table II encompass the performance of both state-of-the-art networks and the proposed network in this study on the GMOD. The best results are highlighted in bold red, while the second-best results are presented in bold black. The proposed GMODet network achieved the most outstanding overall performance, with AP_{50}, AP_{75}, and $AP_{50:95}$ (AP) reaching 65.5%, 48.5%, and 42.3%, respectively. The overall performance of the proposed GMODet network surpasses that of the second-ranked RT-DETR network by 1.9%, 5.1%, and 1.5% in the respective metrics. Furthermore, to evaluate the ability of each model to handle the objects of different scales and their inference speed, we have reported the values of AP_S, AP_M, and AP_L in Table III. Additionally, the network parameters and computational complexity of each network are metrics for assessing model performance and thus are discussed in Table III. Given the constraints of this article's length, we have visualized the top 5 experimental results with the highest AP_{50} scores for both CNN-based models and transformer-based models in their inference outcomes, as shown in Fig. 6. In these visualizations, yellow boxes indicate FPs, red boxes indicate false negatives (FNs), and boxes of other colors represent instances detected in various categories, classified as true positives (TPs).

For CNN-based models, although networks, such as HANet, FSANet, and NWD-RKA, are specifically designed for small object detection, they did not exhibit outstanding performance on the GMOD in this study. We speculate that this is due to the complexity of the object distribution and

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TABLE II

ACCURACY COMPARISON ON THE GMOD TEST SET. THE BEST RESULTS ARE HIGHLIGHTED IN RED FONT. THE NEXT BEST RESULTS ARE SHOWN IN BOLD FACE

Model	Publication	Year	Backbone	AP_{50}				AP_{50}	AP_{75}	$AP_{50:95}$
				MV	MT	CV	FS			
CNNs-based models										
YOLOF	CVPR	2021	R50	71.9	9.8	24.9	33.7	35.1	15.0	17.5
HANet	CSVT	2023	R50	80.7	21.9	32.4	50.3	46.3	24.6	24.9
FSANet	TGRS	2022	R50	78.2	22.0	58.2	42.0	50.1	28.1	27.3
WTHA-ViT	TGRS	2024	WTHA-ViT-S	79.3	23.4	50.8	52.6	51.5	25.1	27.3
TOOD	ICCV	2021	R50	80.4	22.4	51.7	51.7	51.6	29.3	29.5
NWD-RKA	ISPRS	2022	R50	77.8	23.1	62.0	49.7	53.2	26.4	28.3
Faster RCNN	TPAMI	2016	R50	80.8	25.3	59.7	52.0	54.4	29.0	30.5
Sparse RCNN	CVPR	2021	R50	83.0	27.8	61.7	54.5	56.8	33.1	33.3
PPYOLOE-L	Arxiv	2022	CSPRepResNet	86.2	28.7	66.4	60.1	60.3	39.4	35.6
YOLOv8-L	Arxiv	2023	R50	87.4	30.3	**71.9**	55.8	61.4	**45.4**	40.6
Transformer-based models										
DETR	ECCV	2020	R50	59.3	20.9	39.0	34.2	38.3	10.5	16.3
Conditional-DETR	ICCV	2021	R50	59.0	21.3	41.9	44.6	41.7	9.2	15.7
DNTR	TGRS	2024	R50	36.9	8.6	36.3	13.9	47.3	21.1	23.9
DAB-DETR	ICLR	2022	R50	73.8	21.6	56.9	51.9	51.1	14.0	21.6
CO-DETR	ICCV	2023	R50	85.1	30.8	67.3	53.5	59.2	40.3	36.7
DINO-Deformable-DETR	ICLR	2023	R50	85.3	30.1	64.8	58.0	59.6	35.8	35.0
Lite-DETR	CVPR	2023	R50	83.2	31.0	69.3	57.6	60.3	36.5	35.1
RT-DETR	CVPR	2024	R50	**89.3**	**32.8**	66.8	**65.3**	**63.6**	43.4	**40.8**
Ours			R50	**90.0**	**32.6**	**73.3**	**66.0**	**65.5**	**48.5**	**42.3**

TABLE III

STATISTICAL ANALYSIS OF VARIOUS MODELS IN HANDLING OBJECTS OF DIFFERENT SCALES IN THE GMOD, INCLUDING MODEL PARAMETERS, AND PROCESSING EFFICIENCY METRICS. ↑ INDICATES THAT A HIGHER VALUE CORRESPONDS TO BETTER PERFORMANCE, WHILE ↓ INDICATES THAT A LOWER VALUE IS PREFERRED. THE BEST RESULTS ARE HIGHLIGHTED IN RED FONT, AND THE SECOND-BEST RESULTS ARE SHOWN IN BOLD FACE

Methods	AP_S	AP_M	AP_L	Params(M) ↓	GFLOPs↓	FPS↑
YOLOF	4.6	34.1	22.0	42.1	**86.4**	42
HANet	9.0	27.3	34.5	**26.4**	300.7	80
FSANet	12.1	33.5	41.6	**30.7**	154.4	**115**
WTHA-ViT	9.4	31.7	31.8	35.1	171.7	2
TOOD	9.5	34.1	39.1	32.0	159.2	26
NWD-RKA	8.6	31.4	44.8	123.2	249.9	18
Faster RCNN	10.4	35.3	42.3	41.4	197.5	31
Sparse RCNN	17.5	38.2	48.3	106.0	172.7	24
PPYOLOE-L	**29.1**	40.0	44.6	51.3	118.1	73
YOLOv8-L	14.0	**46.2**	51.5	43.6	182.0	74
DETR	2.4	17.0	35.5	41.6	**83.9**	18
Conditional-DETR	3.0	16.9	32.8	43.4	89.2	17
DNTR	8.7	28.0	31.9	128.0	265.6	4
DAB-DETR	6.0	23.9	37.1	43.7	91.3	15
CO-DETR	16.1	41.2	43.4	64.5	738.9	9
DINO-Deformable-DETR	19.1	39.0	43.3	47.5	254.5	31
Lite-DETR	14.1	39.3	52.1	47.8	141.4	64
RT-DETR	21.2	45.0	**57.3**	42.7	150.5	**106**
Ours	**22.4**	**46.7**	**58.0**	42.2	179.5	83

more detailed object categories within the GMOD. Additionally, employing WTHA-ViT-S, which integrates frequency information structures, as the backbone for the WTHA-ViT network combined with the RetinaNet detector enhances global information representation capabilities. However, it fails to demonstrate superior performance on the GMOD. As indicated in Table III through multiscale evaluation metrics, this limitation arises from the inadequate enhancement of global representation capabilities for small objects, with an AP_S of only 9.4%. Faster R-CNN and sparse R-CNN are typical two-stage object detection algorithms. The proposed model demonstrates significantly better computational efficiency and inference speed compared to these algorithms. In comparison to sparse R-CNN, the proposed GMODet network achieves a roughly 4× faster inference speed (83 versus 24 frames/s) and outperforms it by 8.7%, 15.4%, and 9.0% in terms of AP_{50}, AP_{75}, and AP, respectively. Moreover, comparisons with the state-of-the-art single-stage network frameworks, such as YOLOF, TOOD, PPYOLOE-L, and YOLOv8-L, indicate that GMODet achieves a good balance between speed and accuracy. As shown in Table III, PPYOLOE-L and YOLOv8-L demonstrate strong performance in multiscale object detection, with PPYOLOE-L achieving the best small object detection performance ($AP_S = 29.1\%$). However, for large-scale object detection, the AP_L of PPYOLOE-L is 13.4% lower than that of the proposed GMODet network, which is unfavorable

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Fig. 6. Visualization of inference results on the GMOD test dataset. Yellow bounding boxes indicate FPs, red bounding boxes represent FNs, and other colored bounding boxes correspond to detected instances of each category, which are TPs.

for detecting large objects, such as FS. This metric also aligns perfectly with the visualization results shown in Fig. 6. Furthermore, although YOLOv8-L achieves the best detection performance among CNN-based models, it fails to balance multiscale detection capabilities and inference speed, especially for small object detection, with an AP_S of only 14.0%. Thus, YOLOv8-L may not perform well in detecting small objects in complex real-world battlefield scenarios.

In the inference results of transformer-based models, there is a significant disparity in performance metrics across different models. As the first fully end-to-end detection network based on the transformer architecture, DETR, while having relatively low parameter count and computational complexity, exhibits unsatisfactory detection precision. Notably, the AP for small objects (APS) is only 2.4%, indicating a substantial number of missed objects, which is corroborated by the visualization results of DETR in Fig. 6. DNTR, a network specifically designed for small object detection in the remote sensing domain, also failed to demonstrate its advantage on our GMOD. Furthermore, due to the model's enormous parameter

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TABLE IV

ACCURACY COMPARISON ON THE MVRSD TEST SET. THE BEST RESULTS ARE HIGHLIGHTED IN RED FONT. THE NEXT BEST RESULTS ARE SHOWN IN BOLD FACE

Model	Publication	Year	Backbone	AP_{50}					AP_{50}	AP_{75}	$AP_{50:95}$
				SMV	LMV	AFV	MCV	CV			
CNNs-based models											
WTHA-ViT	TGRS	2024	WTHA-ViT-S	60.0	64.4	55.3	4.4	90.8	55.0	42.7	35.6
YOLOF	CVPR	2021	R50	60.4	75.9	80.7	41.5	86.4	69.0	36.5	37.1
Sparse RCNN	CVPR	2021	R50	69.5	74.3	73.7	54.1	86.3	71.6	56.0	47.1
NWD-RKA	ISPRS	2022	R50	74.8	76.6	82.4	48.9	92.1	75.0	53.0	46.4
FSANet	TGRS	2022	R50	74.3	77.8	84.8	47.8	92.5	75.4	52.4	46.6
TOOD	ICCV	2021	R50	79.9	81.7	85.4	46.3	94.9	77.6	58.7	50.2
HANet	CSVT	2023	R50	82.2	84.9	86.7	48.2	94.6	79.3	57.1	49.6
Faster RCNN	TPAMI	2016	R50	79.2	83.5	88.2	53.3	93.4	79.5	59.7	51.1
PPYOLOE-L	Arvix	2022	CSPRepResNet	88.2	88.7	90.6	51.0	97.1	83.1	67.8	56.0
YOLOv8-L	Arvix	2023	R50	91.4	90.1	89.2	63.4	**97.6**	86.3	73.2	60.3
Transformer-based models											
DETR	ECCV	2020	R50	17.8	32.4	32.6	9.2	40.4	26.5	3.1	8.6
Conditional-DETR	ICCV	2021	R50	20.7	34.7	42.4	19.0	46.4	32.6	3.5	10.5
DNTR	TGRS	2024	R50	57.8	62.3	67.2	32.2	85.0	60.9	24.0	29.9
DAB-DETR	ICLR	2022	R50	63.1	72.8	73.2	26.2	83.7	63.8	26.5	31.1
DINO-Deformable-DETR	ICLR	2023	R50	74.4	79.4	83.2	50.7	92.4	76.0	58.1	49.7
Lite-DETR	CVPR	2023	R50	79.0	80.4	86.4	53.5	93.5	78.5	64.1	52.9
CO-DETR	ICCV	2023	R50	78.1	82.0	83.1	63.2	92.8	79.8	63.8	53.0
RT-DETR	CVPR	2024	R50	91.1	**90.1**	**93.4**	**64.7**	97.2	**87.3**	**75.0**	**60.9**
Ours			R50	**92.2**	**89.5**	**93.0**	**69.0**	**97.2**	**88.2**	**75.2**	**61.7**

TABLE V

STATISTICAL ANALYSIS OF THE OBJECT DETECTION CAPABILITY METRICS OF VARIOUS MODELS AT DIFFERENT SCALES ON THE MVRSD. THE BEST RESULTS ARE HIGHLIGHTED IN RED FONT, WHILE THE SECOND-BEST RESULTS ARE PRESENTED IN BOLD

Methods	AP_S	AP_M	AP_L	Methods	AP_S	AP_M	AP_L
WTHA-ViT	23.4	36.8	16.2	YOLOv8-L	**47.8**	61.3	51.8
YOLOF	24.3	38.9	23.0	DETR	5.2	8.9	10.7
Sparse RCNN	35.1	48.6	35.0	Conditional-DETR	3.5	11.2	11.2
NWD-RKA	31.6	47.5	30.2	DNTR	17.1	31.5	12.8
FSANet	37.6	47.2	35.0	DAB-DETR	16.6	33.0	31.9
TOOD	33.9	51.9	26.4	DINO-Deformable-DETR	38.0	50.6	34.4
HANet	32.9	50.9	34.3	Lite-DETR	39.2	54.0	40.5
Faster RCNN	39.7	52.2	32.2	CO-DETR	**45.8**	54.0	37.9
PPYOLOE-L	45.7	56.9	45.0	RT-DETR	43.4	**62.1**	**56.1**
				Ours	44.1	**62.6**	**63.4**

count and computational cost, it is unsuitable for real-time object detection. Similarly, networks, such as conditional-DETR, DAB-DETR, CO-DETR, and DINO-DETR, are not suitable for real-time object detection due to excessive parameter and computational consumption or slow inference speed. Moreover, the aforementioned network structures do not perform well in multiscale object detection scenarios.

Lite-DETR, RT-DETR, and our proposed GMODet network all belong to the category of lightweight or real-time object detection networks. These three networks achieve a balanced performance in terms of AP_{50}, AP_{75}, AP, and FPS metrics. However, Lite-DETR still suffers from missed detections of small objects, with an APS of only 14.1%. As illustrated in Fig. 6, the visualization results of Lite-DETR also demonstrate a significant number of missed detections in dense object scenarios. Although RT-DETR achieves superior detection performance, it still encounters missed detections and FPs in more complex scenarios, such as cases of partial occlusion or when the background is highly similar to the target objects. For example, as shown in columns 3, 4, 5, and 8 of Fig. 6, there are instances of missed detections for MV and FPs for MT. Our proposed GMODet network achieves the best performance on the GMOD. Nevertheless, some shortcomings still exist, as indicated by the yellow rectangles in columns 1 and 6 of Fig. 6, which highlight erroneous detection cases. We hypothesize that these errors are caused by the similarity in features between the objects and the background.

D. Experiments on MVRSD Task

To verify the generalizability of the proposed object detection method, we conducted validation experiments using the MVRSD. The quantitative evaluation metrics in Table IV summarize the prediction performance of all networks on the MVRSD, while Table V presents the performance of each model in multiscale object detection. Fig. 7 visualizes the prediction results of the top 4 models in terms of AP_{50} scores on the test set. Among CNNs-based models, YOLOv8-L achieved the best performance on the MVRSD. Our proposed model demonstrated the highest overall performance across all models. Compared to YOLOv8-L, our model's inference results show improvements of 1.9%, 2.0%, and 1.4% in AP_{50}, AP_{75}, and AP, respectively.

Additionally, compared to the real-time model RT-DETR, our model achieved increases of 0.9%, 0.2%, and 0.8% in the same metrics. Although our model slightly outperformed

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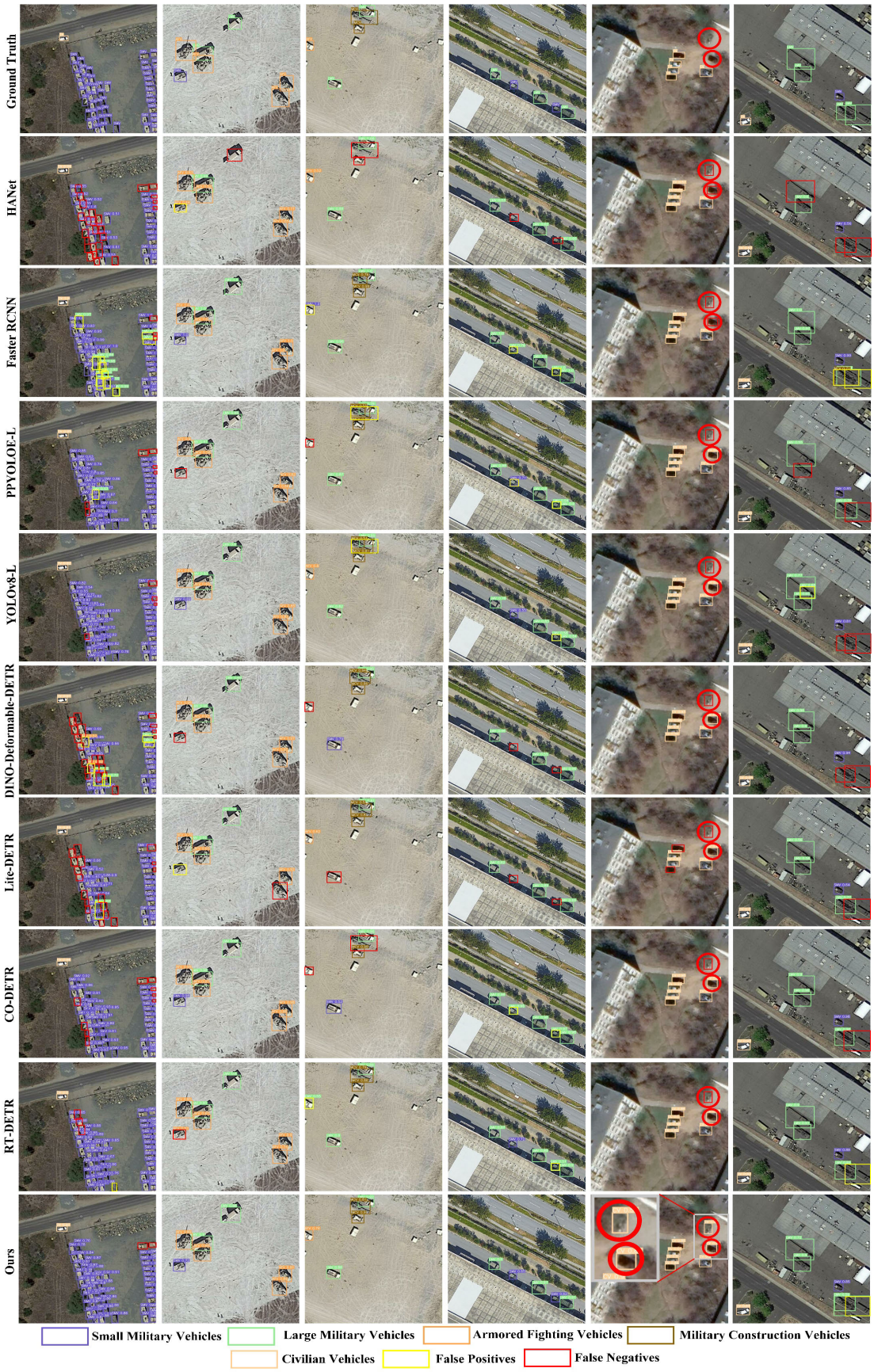


Fig. 7. Visualization of the inference results for the MVRSD test dataset. The yellow boxes indicate FPs, the red boxes indicate FNs, and the boxes in other colors represent instances of each category detected as TPs.

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TABLE VI
ABLATION EXPERIMENT ON THE GMOD. THE BEST RESULTS ARE IN BOLD. ↑ INDICATES THAT A HIGHER VALUE CORRESPONDS TO BETTER PERFORMANCE, WHILE ↓ INDICATES THAT A LOWER VALUE IS PREFERRED

Baseline	RAW	CCFA	SPIM	AP_{50}	AP_{75}	$AP_{50:95}$	AP_S	AP_M	AP_L	Params (M)↓	GFLOPs↓	FPS↑
√				62.3	45.1	40.4	23.0	44.8	50.2	**31.7**	**97.7**	**134**
√	√			63.6	46.5	41.4	**23.3**	45.2	54.7	40.5	106.0	120
√	√	√		63.8	48.0	41.4	21.9	45.7	56.4	40.6	132.1	93
√	√	√	√	**65.5**	**48.5**	**42.3**	22.4	**46.7**	**58.0**	42.2	179.5	83

RT-DETR in overall performance, the multiscale prediction results presented in Table V indicate that our model showed significant improvements in AP_S, AP_M, and AP_L, particularly in large-scale object detection, where AP_L increased by 7.3%.

Based on the analysis of Tables IV and V and Fig. 7, CNNs-based models and transformer-based models demonstrate comparable overall performance on the MVRSD. However, CNNs-based models exhibit superior performance in small object detection, with YOLOv8-L achieving the best results among all models, reaching an AP_S of 47.8%. Moreover, WTHA-ViT, utilizing WTHA-ViT-S as its backbone, achieved the poorest performance, primarily due to its failure to detect large-scale objects, as evidenced by an AP_L of only 16.2%. We hypothesize that this is attributable to the structural complexity of objects, such as LMV, AFV, and MCV, which are often composed of multiple substructures forming intricate object entities. This complexity likely exceeds the network's global feature extraction capabilities, resulting in suboptimal performance on this dataset and leading to missed detections of large-scale objects. In contrast, transformer-based models encounter more missed detections, particularly in scenarios involving dense or small objects. For instance, as illustrated in the first column of Fig. 7, transformer-based models show limited competitiveness in detecting SMV objects. Furthermore, as indicated in Table V, detection, with the DETR and conditional-DETR models, achieving AP_S values of only 5.2% and 3.5%, respectively. Nonetheless, transformer-based models excel in medium and large object detection, with RT-DETR and our proposed model ranking among the top two performers for these scales.

Moreover, the MVRSD primarily highlights the challenges of detecting densely packed objects and recognizing objects in the presence of background blur. Columns 4 and 5 in Fig. 7 present the examples of detection results under conditions where shadows obscure the objects or when objects are hidden in the background. Notably, the predicted results circled in red in Column 5 illustrate this point. These objects were overlooked in the original dataset annotations, but our model, as well as other competitive transformer-based models, was able to predict them. Specifically, our model successfully detected CV objects camouflaged in dense vegetation. However, our model also exhibited some limitations in the MVRSD, as shown in the yellow rectangle in the last column of the result visualization in Fig. 7. Although our model successfully predicted the presence of LMV objects, the bounding box localization was inaccurate, resulting in the objects being merged with adjacent LMV objects.

E. Ablation Study

To validate the inference effectiveness of the various modules proposed in Section III of this study, we conducted a series of ablation experiments. The performance of different modules under ablation testing is presented in Table VI. Our baseline network is derived from the architecture shown in Fig. 1, with the encoder section removed, resulting in a baseline of "Backbone (ResNet50) + Decoder." As indicated in Table VI, both the parameter count and computational load of the baseline network are the lowest among the configurations tested.

Building on this foundation, we incorporated the RAW structure into the encoder section. Although this integration significantly increased the overall parameter count and computational complexity of the network, the prediction results improved by 1.2%, 1.4%, and 1.0% in terms of AP_{50}, AP_{75}, and AP, respectively, compared to the baseline network. This improvement is deemed worthwhile. Furthermore, after incorporating the CCFA structure, the model shows significant improvements across multiple metrics. Compared to the model using only the "Baseline + RAW" combination, the inclusion of CCFA results in an increase in key performance indicators, such as AP_{50} and AP_{75}, with improvements of 0.2% and 1.5%, respectively. Notably, the performance enhancement for large (AP_L) and medium (AP_M) objects reaches 1.7 percentage points, indicating that the CCFA structure plays a positive role in enhancing the detection capabilities for large and medium-sized objects. Although the performance for small objects (AP_S) slightly decreases, the overall performance of the model still improves, with the model's parameters (Params) and computational cost (GFLOPs) remaining within a reasonable range, with almost no increase in parameter count. Finally, the introduction of the SPIM structure led to a notable enhancement in the model's prediction results, with increases of 1.7%, 0.5%, and 0.9% in AP_{50}, AP_{75}, and AP, respectively, compared to the previous experimental setup. While the addition of SPIM resulted in a higher parameter count and computational load, the continuous integration of coarse semantic information and spatial data yielded a substantial overall improvement in detection performance. We believe this tradeoff is justified, as it achieves a balance between accuracy and inference speed (with an FPS rate of 83).

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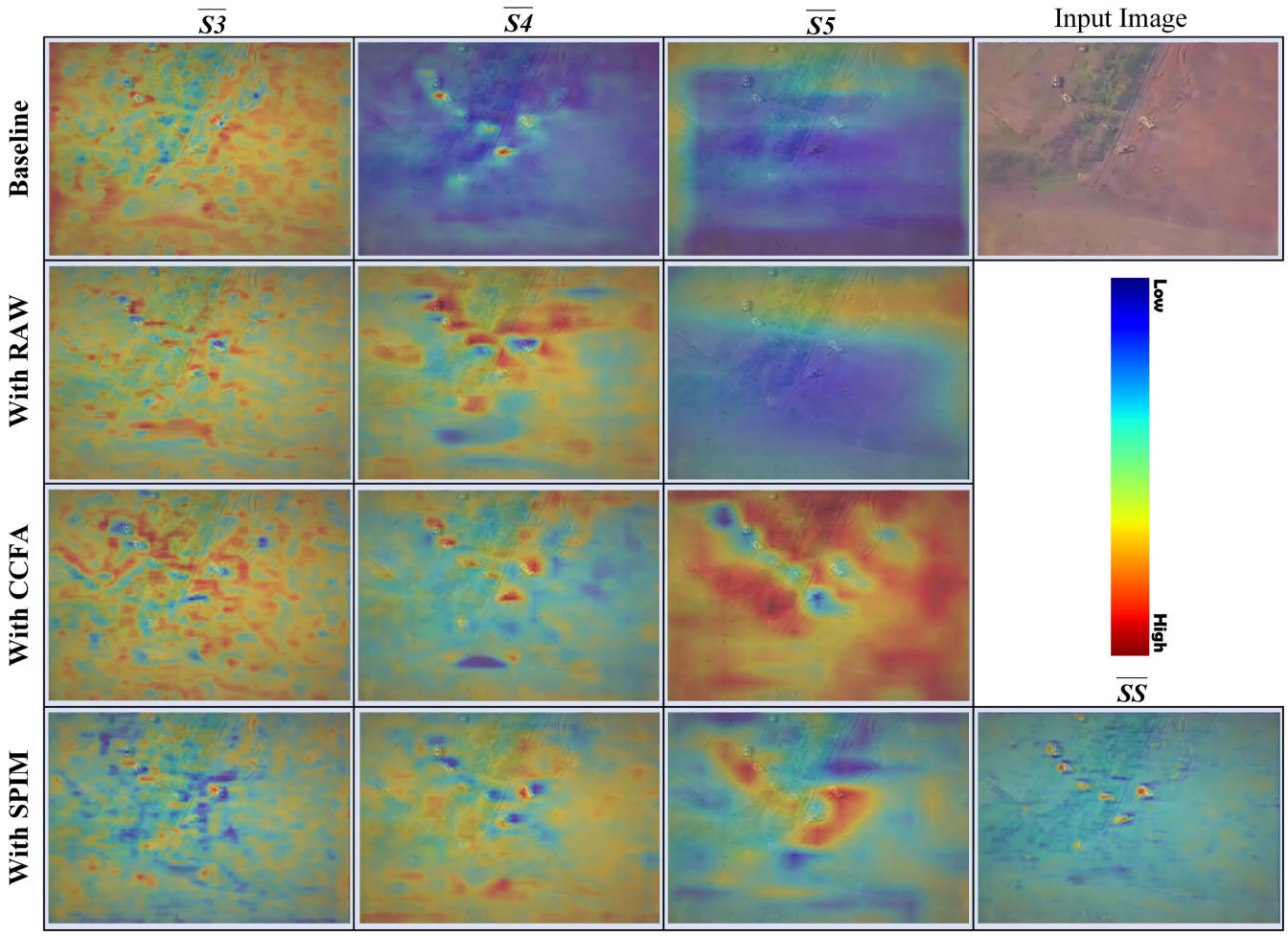


Fig. 8. Visualization of intermediate layer features under different modules. The rows show the visualization results of feature maps of different modules; the columns show the output of the last three layers of features $\overline{S3}$, $\overline{S4}$, and $\overline{S5}$ in the encoder stage. Our model outputs the semantic–spatial feature $\overline{SS}$ in the encoder stage.

In order to intuitively interpret the enhanced performance exerted by each module, we visualize the last three layers of features $\overline{S3} - \overline{S5}$ output by the encoder in Fig. 8. The model we proposed has an additional layer of $\overline{SS}$ feature maps of semantic–spatial information. As shown in the first row of Fig. 8, when using the baseline network for inference, the network structure does not significantly express the characteristics of the object, especially the first layer $\overline{S3}$, which was enhanced at the same time as the background features. Although the $\overline{S4}$ feature pays some attention to objects, it pays less attention to objects with obscure features, such as vehicles hidden in trees. The $\overline{S5}$ feature is a high-level abstract feature, but it still does not show significant feature expression. With the addition of the RAW structure, the advanced features $\overline{S3} - \overline{S5}$ become more focused on the detection of vehicle objects. This aligns with the quantitative results presented in Table VI, which significantly enhances the attention to multiscale objects, particularly small objects. After incorporating RAW, the AP_S achieved the best performance across all experiments, reaching 23.3%. As shown in Fig. 8, the configuration of the CCFA effectively enhances the differentiation between object and background features by reducing the impact of background features, allowing $\overline{S3} - \overline{S5}$ to continually focus on the object features. In conjunction with the data presented in Table VI, the introduction of the CCFA significantly improves the model's detection accuracy, particularly highlighting the importance of the fused features in enhancing the discriminability of complex, larger objects. Furthermore, SPIM significantly improves the model's discriminative capability regarding objects, greatly diminishing the impact of background interference, particularly in the semantic–spatial feature map $\overline{SS}$, thereby allowing the model to concentrate on expressing vehicle's features more effectively.

V. Discussion

A. Effectiveness of Different Components in RAW

1) Validity Analysis of Components in RAW: To validate the roles of the three small components proposed in RAW—GRAU, LRAU, and DSAU—we conducted an ablation study on the effectiveness of these components while keeping the other structures of the GMODet network unchanged. As shown in Table VII, when only the GRAU structure is integrated,

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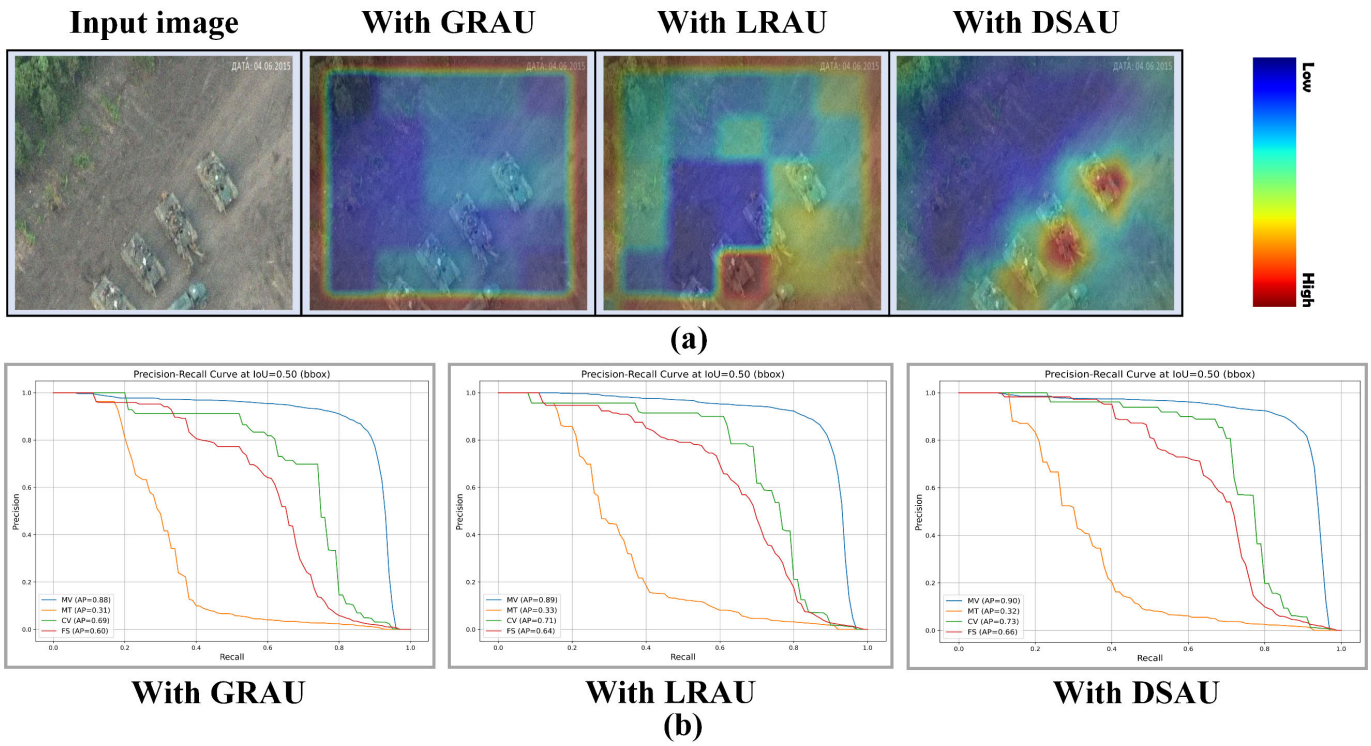


Fig. 9. Analysis of the effectiveness of different components on the network in RAW. We visualized (a) intermediate layer feature maps and (b) P-R curves under different component combinations.

TABLE VII

ABLATION EXPERIMENT ANALYSIS OF THE EFFECTIVENESS OF COMPONENTS IN RAW. THE BEST RESULTS ARE IN BOLD

GRAU	LRAU	DSAU	AP_{50}	AP_{75}	$AP_{50:95}$	AP_S	AP_M	AP_L
√			62.0	42.6	39.8	23.2	44.4	56.6
√	√		64.0	45.0	41.1	**23.7**	44.5	57.1
√	√	√	**65.5**	**48.5**	**42.3**	22.4	**46.7**	**58.0**

TABLE VIII

SENSITIVITY ANALYSIS OF WINDOW SIZE W_s VALUE TO NETWORK PERFORMANCE. THE BEST RESULTS ARE IN BOLD

Scheme	AP_{50}	AP_{75}	$AP_{50:95}$	AP_S	AP_M	AP_L
$W_s = 1$	64.0	46.2	41.5	22.5	45.3	54.1
$W_s = 3$	**65.5**	**48.5**	**42.3**	22.4	**46.7**	58.0
$W_s = 5$	64.5	46.1	42.1	**24.5**	45.1	59.2
$W_s = 7$	63.5	46.6	41.5	20.9	45.4	57.2
$W_s = 9$	63.6	46.2	41.7	23.2	45.1	**59.8**
$W_s = 12$	63.1	47.2	41.5	23.2	45.8	54.2

the overall performance is suboptimal, with AP_{50}, AP_{75}, and AP values of only 62.0%, 42.6%, and 39.8%, respectively. We believe that considering only global information is insufficient, as it may lead to inaccuracies in boundary localization for larger scale objects, resulting in decreased detection accuracy. This observation is consistent with the intermediate feature map representation in Fig. 9(a), where it can be seen that integrating only the GRAU structure enhances both the object domain and the background simultaneously.

The introduction of the LRAU structure resulted in an overall performance enhancement, with improvements in AP_{50}, AP_{75}, and AP values of 2.0%, 2.4%, and 1.3%, respectively, compared to the configuration that only incorporated the GRAU structure. The intermediate layer feature maps depicted in Fig. 9(a) exhibit a more focused representation of object domain features. Therefore, it is essential to consider both global and local information to improve object recognition rates, as this enhances the boundary localization capability for larger objects. Furthermore, the addition of the DSAU structure, in conjunction with the three modules, led to a further performance boost, with AP_{50}, AP_{75}, and AP values increasing by 1.5%, 3.5%, and 1.2%, respectively, compared to the previous experimental round. Detection of multiscale objects has also become more balanced, as evidenced by the AP_S, AP_M, and AP_L values. We hypothesize that the DSAU structure provides enhanced spatial attention to multiscale objects. Additionally, the P–R curves visualized in Fig. 9(b) indicate that the combined regional perception capabilities of the three modules are stronger, reflected in an improved detection capability for larger objects, particularly in the CV and FS categories. Notably, the AP_{50} value for FS increased by approximately 6.0% and 2.0% compared to the results of the first two experimental setups.

2) Analysis of Window Size in RAW: The setting of window size W_s in the RAW structure has a certain impact on model performance. As shown in the statistical analysis of the sensitivity of network performance to the window size W_s in Table VIII, we selected various window sizes, such as 1, 3,

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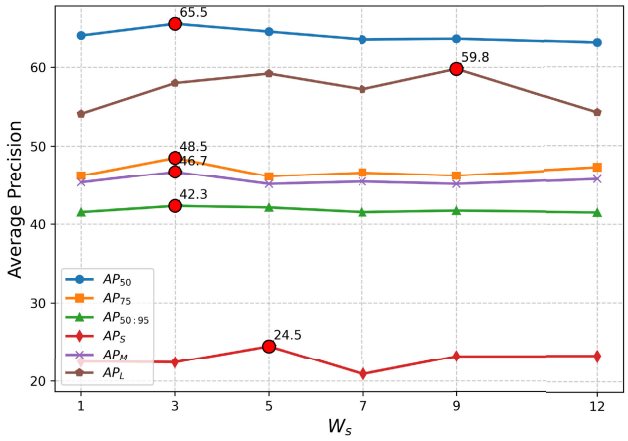


Fig. 10. Sensitivity analysis of the W_s value of RAW on the performance of the GMODet network. The red dot represents the best result.

5, 7, 9, and 12 for verification experiments. A higher value of W_s indicates a larger window, which accommodates more objects and increases the likelihood of detecting large-scale objects. As illustrated in Fig. 10, when W_s is set to 3, the model achieved the best prediction results, with evaluation metrics reaching their highest overall level. According to the statistical analysis of the dataset in Section IV-A, the minimum pixel area of the object bounding box is 12. Therefore, we hypothesize that when W_s is set to 3, it retains the global features of the smallest objects within the window while also considering the local features of larger objects from a distance. Conversely, when W_s is set to a higher value, the ability of adjacent windows to express the global features of small objects diminishes, leading to a slight reduction in small-scale object detection capabilities. However, the detection ability for medium- and large-scale objects improves. On the other hand, if W_s is set to a smaller value, it increases the detection of background noise, resulting in poor detection performance. Based on the comprehensive statistical results presented in Table VIII, a window size of 3 is deemed the most suitable. This setting achieves a balanced detection of both small and larger scale objects.

B. Effectiveness of CWA

To validate the effectiveness of CWA in the CCFA, we replaced CWA with "concatenate" and conducted comparative validation experiments. Table IX compares the performance metrics of the two approaches (concatenate and CWA) in the GMODet network for ground-moving object detection tasks. It is evident that CWA outperforms the concatenate method, achieving increases of 0.5%, 1.1%, and 0.2% in AP_{50}, AP_{75}, and AP values, respectively. This indicates that CWA has superior performance and provides more robust detection results. Furthermore, although the concatenate method shows better AP_S compared to CWA, CWA demonstrates more impressive performance in medium-scale and large-scale object detection. In particular, for large-scale object detection, CWA improves AP_L by 4.6% over the concatenate

TABLE IX

EFFECTIVENESS ANALYSIS OF CWA ON NETWORK PERFORMANCE. THE BEST RESULTS ARE IN BOLD

Scheme	AP_{50}	AP_{75}	$AP_{50:95}$	AP_S	AP_M	AP_L
Concatenate	65.0	47.4	42.1	**24.6**	46.2	53.4
CWA	**65.5**	**48.5**	**42.3**	22.4	**46.7**	**58.0**

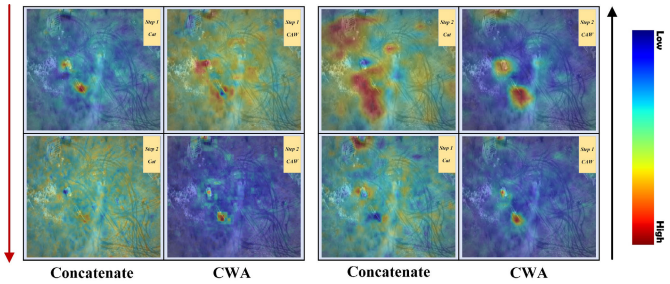


Fig. 11. Visualization of intermediate layer feature maps corresponding to different feature fusion methods. The results of top–down fusion and bottom–up fusion in the CCFA module are visualized separately.

TABLE X

EFFECTIVENESS ANALYSIS OF SIEU ON NETWORK PERFORMANCE. THE BEST RESULTS ARE IN BOLD

Scheme	AP_{50}	AP_{75}	$AP_{50:95}$	AP_S	AP_M	AP_L
Without SIEU	63.7	44.9	40.9	**23.8**	44.8	55.3
SIEU	**65.5**	**48.5**	**42.3**	22.4	**46.7**	**58.0**

method. This aligns with the visualized results of multiple fusions in CCFA, as shown in Fig. 11. Fig. 11 visualizes the results of top–down and bottom–up fusion in CCFA. Additionally, Fig. 11 indicates that the CWA fusion approach is more focused on the global features of multiscale objects, leading to less feature enhancement of complex background contents.

Overall, we believe that CWA possesses a competitive advantage in multiscale feature fusion, making it more suitable for integrating multiscale features in complex scenarios. We hypothesize that CWA enhances more global scale information by leveraging frequency knowledge, which is more favorable for larger objects.

C. Effectiveness of SIEU

To verify the effectiveness of the SIEU shown in Fig. 4, we conducted experiments by replacing the SIEU structure, and the statistical results of the experimental outcomes are presented in Table X. In the experiments, we replaced the SIEU structure with operations, such as "Maxpool-Conv-Concatenate" for feature fusion along the channel dimension. Specifically, as illustrated in Fig. 4, the spatial features ***S1*** and ***S2***, along with the coarse semantic features $\overline{\boldsymbol{S3}}^3$, underwent channelwise feature fusion, with the prerequisite that the sizes of all features were unified through bilinear interpolation.

The quantitative metrics presented in Table X indicate that structures that do not utilize SIEU for semantic–spatial feature coupling exhibit higher detection accuracy for small-scale objects. However, considering the overall detection capability

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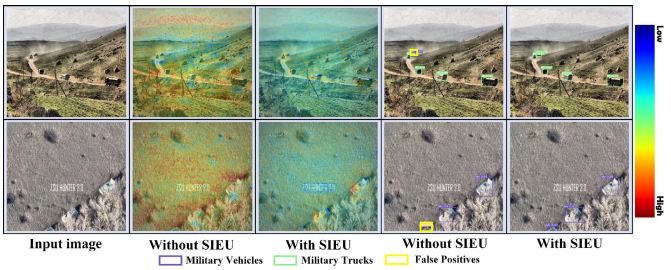


Fig. 12. Analysis of the effectiveness of SIEU on network performance. We visualize the intermediate layer feature maps and the predicted results. Columns 2 and 3 display the visualization of feature maps; columns 4 and 5 present the prediction results.

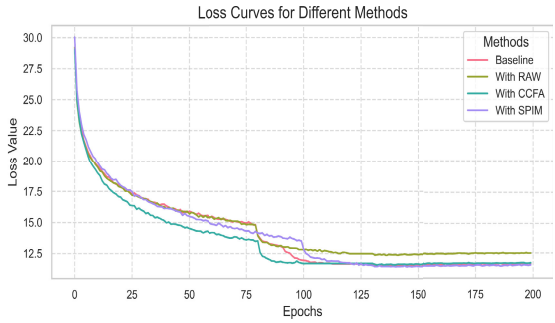


Fig. 13. Visualization of convergence during the training process under different methods.

across multiple scales, the SIEU we designed demonstrates superior performance in semantic–spatial integration in complex scenes. As shown in Fig. 12, structures lacking SIEU for semantic–spatial feature coupling tend to focus more on the background, leading to decreased detection accuracy for distant objects under oblique imaging conditions and partially occluded objects, as well as an increased FP rate. In contrast, the structure of SIEU enables the overall network weights to focus more on the object, allowing for better differentiation between the background and the object, as well as distinguishing the differences between the objects of different categories, thereby enhancing detection accuracy.

D. Convergence Analysis of Training Process

As shown in Fig. 13, we present the loss convergence under different training methods, including the baseline, and visualizations of the loss functions incorporating RAW, CCFA, and SPIM. Although the SPIM approach exhibits a slower convergence rate during the initial training phase, it ultimately surpasses the other three methods in terms of convergence as training progresses. This phenomenon is primarily attributed to the introduction of the cross-entropy loss function in SPIM, which facilitates a deeper exploration of coarse-grained semantic information, thereby increasing the difficulty in the early stages of training. However, this complexity also enables the model to better learn the underlying features of the data, guiding the learning of coarse semantics, and resulting in improved performance in later stages. Overall, the various strategies we propose demonstrate stable convergence on the GMOD and exhibit strong inference capabilities, effectively validating the efficacy of the proposed strategies in enhancing model robustness and generalization abilities.

E. Real-Time Inference Capabilities for Large-Format Imagery

The test set of the MVRSD contains three large-scale satellite images of complex scenes, with the largest image having a resolution of up to 9152 × 9152 pixels. These images are not manually annotated. To evaluate the proposed method's inference capability and generalization performance in large-scale scenarios, we conducted large-scale inference on these three images. Specifically, to prevent hardware limitations caused by directly inputting large-scale images, or extensive missed detections when downscaling the large images to predict on smaller scales, each large-scale image was cropped into multiple smaller images of 672 × 672 pixels for prediction. During inference, the overlap ratio between neighboring small images was approximately 60%, and the predicted results were stitched back into the original large-scale images. The inference time for the largest 9152 × 9152 pixels image was 25 s. In practice, a lower overlap ratio results in faster inference speed, but for a balance between speed and accuracy, an overlap ratio of 60% was ultimately chosen in this study. As shown in Fig. 14, the visualization results of the three large-scale satellite images demonstrate that the red rectangles highlight the enlarged, clear details of local regions. The visualizations intuitively validate the superior object detection and inference performance of our model in complex large-scale scenes, showcasing its robustness and generalization capabilities. Notably, our model maintained stable detection performance in identifying various fine-grained ground vehicles, even in dense and complex scenes. Future research will continue to explore model lightweighting and aim to deploy our model on satellites for on-orbit intelligent object detection.

F. Limitations

The GMODet network proposed in this study has undergone extensive experimental validation on the GMOD and MVRSD in real-world complex scenarios, demonstrating excellent robustness and generalization performance. However, the GMODet network also exhibits certain limitations, which are briefly described in Section IV-C, and highlighted within a yellow rectangle. Our model encounters FPs when objects are highly similar to other background features, particularly evident in cases of building misdetection or noise misdetection. As shown in Fig. 15, the first column illustrates instances of building misdetection, revealing that the images are captured using oblique photography. Notably, the buildings at the bottom have shapes and colors strikingly similar to those of MTs, leading to false detections during inference. Additionally, FPs are present in certain noise contexts, such as the noise points depicted in the second column. Analysis suggests that this noise may pertain to localization markers from drone photography; however, due to their resemblance to MVs, they are misdetected. Overall, despite the occurrence of FPs, minimizing missed detections remains the primary objective in practical applications. In future research, we will continue to optimize our model architecture to further reduce the FP rate and enhance the focus on extremely small objects.

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Fig. 14. Inference results of large-scale satellite imagery in complex scenes. The red rectangular boxes indicate magnified and clearer visualization results.



Fig. 15. Cases of failure in existence in our model. The FPs are indicated as highlighted by the yellow rectangle in the figure.

VI. Conclusion

This study summarizes the current state of research on ground-moving object detection and identifies significant challenges that persist in this field. Therefore, this study focuses on the discovery of ground-moving objects in complex scenarios and proposes a real-time detection method for ground-moving objects in optical remote sensing images, called the GMODet network. This network framework integrates key feature enhancement modules, such as RAW, CCFA, and SPIM, whose interactions enhance the focus on discriminative features for fine-grained detection of ground-moving objects in complex environments. Extensive validation experiments conducted on two different types of datasets demonstrate that our proposed method exhibits exceptional robustness and generalization in fine-grained object detection. Our model achieved the best overall performance among all models evaluated on the GMOD, with scores of 65.5%, 48.5%, and 42.3% for AP_{50}, AP_{75}, and AP, respectively. Furthermore, in the validation of the MVRSD, AP_{50}, AP_{75}, and AP reached 88.2%, 75.2%, and 61.7%, respectively, achieving new heights in fine-grained ground-moving object detection. Notably, we validated the detection task using large-scale satellite imagery in complex scenarios, achieving a balance between detection accuracy and real-time speed, which can assist in expanding the detection of ground-moving objects in conflict areas and enable large-scale safety and environmental threat assessments.

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Nevertheless, our model has certain limitations, particularly in cases where the target objects closely resemble other background features, leading to false detections. In future work, we will place greater emphasis on improving the model's fault tolerance to reduce the probability of false detections. Additionally, we aim to encapsulate the model for deployment on satellites to execute real-time intelligent object detection and result inference on-orbit using satellite imagery.

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